

Q35762

Fax # 580-765-0182

Date _____

7/22/2026

Consumers Energy
Miller Trucking
1800 Mitchell St
Jackson, MI 49203
Attn: Austin Owen

P.O. No.	Ship	Via	FOB
260128-2	7/22/2026	Cust Arranged	Ponca City OK

Quantity	Item Code	Description
2	01000000000	8" 900# RFFE Grove B5 - Gear Op with HW, per 73700sp with inspection Item #260126-2-1 SN# 122352167001, 122432354002 Paper tag: PO#: 4501026629 GS-00505 45063186 OVS-SAL LAT3 REPL

QuarterTurn

RESOURCES INC.

P.O. Box 1455

Ponca City, OK 74602

Office: (580) 765-0183

Fax: (580) 765-0182



CERTIFICATE OF CONFORMANCE

QTR WO#: Q35762-1

Customer: Energy Equipment - Consumers

PO#: 260128-2

Ref PO# 4501026629

Description: 8" 900# RFFE Grove B5, G/O

Quantity: 2

S/N(s): 122352167001, 122432354002

THE EQUIPMENT LISTED ABOVE CONFORMS TO THE FOLLOWING:

B16.34: ☒

NACE MR-0175; ☒
Part 1 & 2:

Fire-Safe per ☒
API 6FA:

Pressure Balance / Communication Hole: Yes: ☐ No: ☒

API 6D 23rd Ed: ☐

API 6D 24th Ed: ☒

API 6D 25th Ed: ☒

We certify that the material listed above has been manufactured and tested in accordance with the current revisions listed above (edition, errata, and addenda), Valvitalia standards, and customer purchase order requirements. Material listed has successfully passed all mandatory test and meets supplemental requirements as noted above.

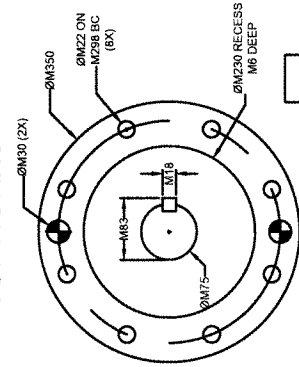
Michael Carson

Quality Assurance Manager

Date: 7/17/2026

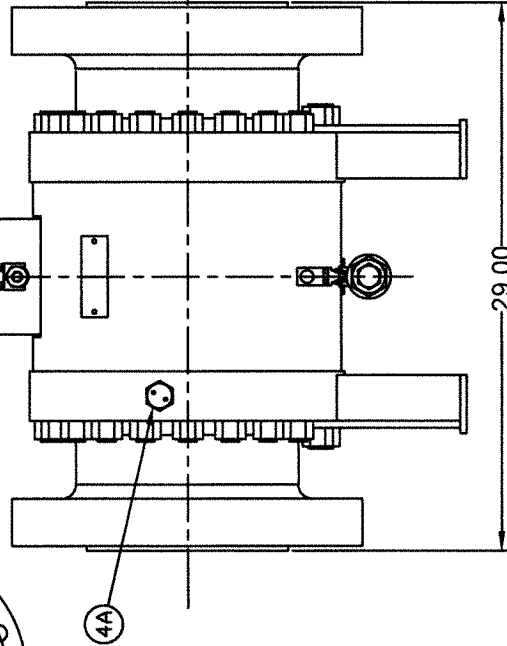
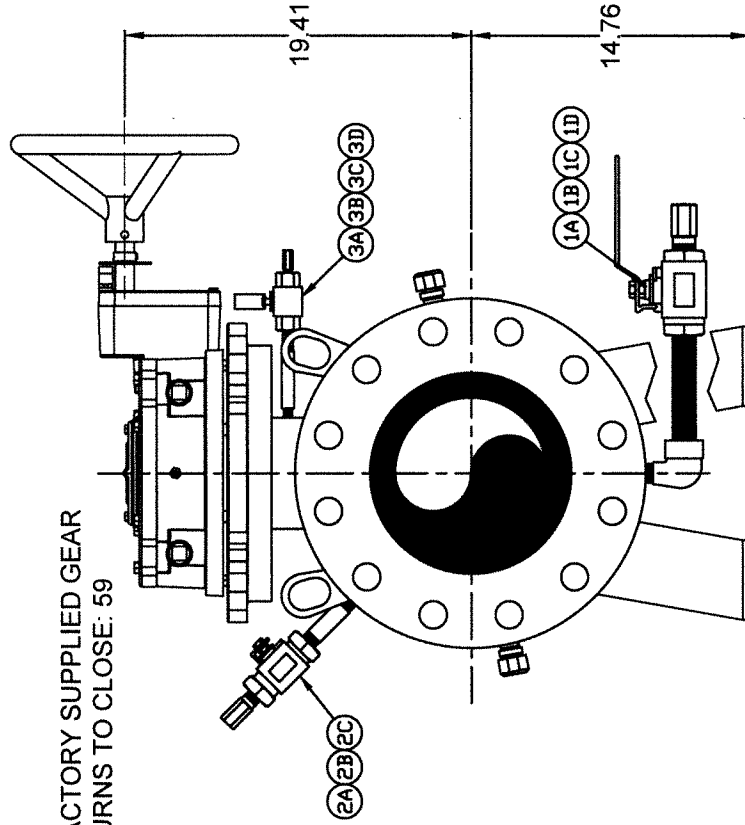
DRAIN ASSEMBLY			RELIEF ASSEMBLY			STEM LUBE ASSEMBLY			SEAT LUBE ASSEMBLY		
#	QTY	DESCRIPTION	MATERIAL	#	QTY	DESCRIPTION	MATERIAL	#	QTY	DESCRIPTION	MATERIAL
1A	1	1" 90° 3M STREET ELBOW	A105	2A	1	1/2" SCH. 80 PIPE NIPPLE	A106B	3A	1	1/4" STEM FITTING	A350 LF2
1B	1	1" SCH. 80 PIPE NIPPLE	A106B	2B	1	1/2" 3000# BV (PER SPEC.)	CF8M	3B	1	1/4" SCH. 160 PIPE NIPPLE	A106B
1C	1	1" 3000# BV (PER SPEC)	CF8M	2C	1	1/2" RELIEF VALVE	F316L	3C	1	1/4" 6000# BV (PER SPEC.)	CF8M
1D	1	1" BULL PLUG	A234 WPB		1	SS TAG: UPPER BODY CAVITY	SS	3D	1	1/4" BULL PLUG	A234 WPB
	1	SS TAG: LOWER BODY CAVITY DRAIN	SS		1	VENT			1	SS TAG: EMERGENCY USE ONLY	SS

TOPWORKS



Ø12.00

FACTORY SUPPLIED GEAR
TURNS TO CLOSE: 59



STEM TORQUE

BTO: 20,760 IN LBS

RUN: 3,912 IN LBS

MAST: 144,000 IN LBS

RIMPULL

BTO: 59LBS

RUN: 13LBS

MAX OUTPUT: 90,000IN LB

MAX INPUT: 300LBS

FLOW COEFFICIENT Cv: 8,531 USGPM

REV. 1 - ADDED SS TAGS, TOPWORKS, LIFT EYES, WEIGHT, CORRECTED MAOP, POF, CONFIRMED DIMS PER MFG SUPPLIED DWG, AND REMOVED BV FROM SEAT LUBE (TT - 4/2/26)
REV. 2 - CORRECTED FITTING SIZES (TT - 9/15/26)

TAG: PO# 4501026629
LINE ITEMS 1 & 2

ALL DIMENSIONS IN INCHES

REV 2

CUSTOMER:	ENERGY EQUIPMENT
PO#:	260128-2 (ITEM# 260126-2-1)
	BUILT AND TESTED PER API 6D & CE 73700SP
	MEETS NACE MR0175 REQUIREMENTS
VALVE MAOP:	2220PSI @-50/100°F, 350°F MAX
BORE:	THRU 7.91"
SEATS:	DIB-1
WEIGHT:	TOTAL APPROX. 1740LBS
COATING:	GOM WM 11W-128 (CONSUMER GREEN)

DWG#:	B0892-010_Q35762
DRAWN BY:	TT
CHECKED BY:	KRB
DATE:	3/6/26
SCALE:	N/A
DIMENSIONAL DRAWING	
GROVE 8" 900 RFFE B5	
FACTORY SUPPLIED G/O	





Shandong Cameron Petroleum Equipment Pte Ltd
No. 2677 Gaodun 8th Road, Gaodun District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0801

CERTIFICATION OF COMPLIANCE API 6D 25TH EDITION

Customer Name: SCHLUMBERGER CANADA LIMITED
CC STO No./PO No. 4710718222
Sales Order Reference Number: N/A
Item Number: 30
Work order Number: 122352187
Quantity of Valves: 2

We hereby certify that the materials for component parts and the manufacture of said valves are in compliance with Cameron valves specifications and drawings and meet all requirements of the above referenced customer order.

We further certify that the above listed valve(s) are in full conformance with American Petroleum Institute (API) 6D 25th Edition and all addenda (as of this date hereby certified), have been API monogrammed and have/have successfully passed all applicable test in accordance with requirements thereof, including additional customer requirements agreed upon in the above indicated order.

ID	Serial Number	Tag No.	Valve Part Number	Valve Type	Valve Size	Pressure Class	End Conn.	Trim	Code of pressure-temperature rating	Balance hole	Design code of pressure boundary bolting	Design code of pressure containing parts
1	122362167001	N/A	2510873-08-06-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim	ASME B16.34	Yes	EN12518-2	ASME VIII Div 1
2	122362167002	N/A	2510873-08-06-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim	ASME B16.34	Yes	EN12518-2	ASME VIII Div 1

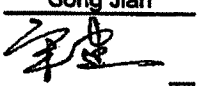
**WCH 122362170 used to modify Ser# 001 to new PTF 2510873-08-06-01

**WCH 122378059 used to modify Ser# 002 to new PTF 2510873-08-06-01

Signature:

Name: Song Jian

Date: 2025.6.24

	EU Declaration of Conformity Pressure Equipment Directive 2014/68/EU		Doc.No.: SLB-QF-A344-093-1	
			Revision: 14	
	Product Category III Conformity Assessment Module H, used for group I.		Customer Name:	
			SCHLUMBERGER CANADA LIMITED	
		PO number: 4710718222		
		Sales Order Number: N/A		
Notified Body	Manufacturer			
LRQA Nederland B.V. at George Hintzenweg 77, 3068 AX Rotterdam, The Netherlands. Identification No.0343 Certification No. 0343/PED/0038/PED/MUM/071 0044/1	Shangdong Cameron Petroleum Equipment PTE Ltd. No. 2577 Gaoxin 5th Road, Gaoxin District. Weifang City, Shandong Province, P.R. China,261041 Tel:86-536-2080601 Tax:86- 536-2080617			
We hereby declare that product(s) described below are in conformance with the pressure Equipment Directive 2014/68/EU This declaration of conformity is issued under the sole responsibility of the manufacturer				
Product Model Number	Qty	Serial Number	Technical File No.	Rev.
Grove B5	1	122352167001	B5 Technical file	1
Grove B5	1	122352167002	B5 Technical file	1
Other EC Directives: _____ Ref. Design Standards: ASME Section VIII Div I, Section II Part D:API 6D, Valves Asme16.34, Valve-Flanged, Threaded, Welding Ends.				
QA Representative: <u>Song Jian</u>		Title: <u>QA Manager</u>		
Signature: <u></u>		Date: <u>2025.6.24</u>		



Shandong Cameron Petroleum Equipment Pte Ltd
No. 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0601

CERTIFICATION OF COMPLIANCE NACE MR0175/ISO15156

Customer Name:
CC STO No./PO No.
Sales Order Reference Number:
Work order number:

SCHLUMBERGER CANADA LIMITED
4710718222
N#A
122352167

Cameron certifies the subject product in compliance with pre-qualified materials of NACE MR0175/ISO15156. This certification does not imply or warrant the application of the product in compliance with NACE MR0175/ISO15156 service conditions in which the customer/user installs the product.

ID	Serial Number	Valve Part Number	Valve Type	Valve Size	Pressure Class	End Conn.	Trim
1	122352167001	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim
2	122352167002	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim

Signature: _____

Name: Song Jian

Date: 2025.6.24



Shandong Cameron Petroleum Equipment Pte Ltd
No. 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0601

LETTER OF COMPLIANCE For:
GROVE BRAND TRUNNION VALVES
Model B4, B5,B7-Sour Low Temperature Service

Customer Name:
CC STO No./PO No.
Sales Order Reference Number:
Item Number:
Work order number:

SCHLUMBERGER CANADA LIMITED
4710718222
N/A
30
122352167

Cameron Valves & Process Systems hereby certifies that the design, manufacture, sampling, testing and quality control inspection of the GROVE brand CSA Z245.15-22 ball valves for Sour Low Temperature Service meet the requirements of CSA Z245.15 Sour Service, VPS Engineering Standards, and customer purchase order requirements. As part of compliance to these specifications, the valves are marked in accordance with MSS SP-25. As part of conformance to these specifications, the valves are marked in accordance with the requirements of API 6D and CSA Z245.15:22 Table 1 and are suitable for installation in CSA Z662 pipeline systems."

The materials of construction for these valves conform to the metallurgical requirements of NACE MR0175/ISO1516 (Parts 1 & 2 for Carbon & Alloy Steels in SSC Region 3 Sour Service and Parts 1 & 3 for CRA's as specified in CSA Z245.15 Chapter 10). According to NACE MR0175/ISO 15156, it is the customer's / user's responsibility to ensure that the valve materials will be satisfactory in the intended service.

The minimum service temperature of these valves is -46°C/50°F. Impact testing has been performed in accordance with applicable ASTM requirements and those of CSA Z245.15 for Category II -45°C Service.

The ASTM A320 Grade L7M studs and cap screws and ASTM Grade A194 7ML nuts used in the products specified above have been Quality Control inspected and meet the requirements of the applicable ASTM specifications-as is already implied fully by the statements above regarding both CSA and NACE standards compliance.

ID	Serial Number	Valve Part Number	Valve Type	Valve Size	Pressure Class	End Conn.	Trim
1	122352167001	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim
2	122352167002	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim

Approved by: _____

Song Jian
Quality manager

Date: 2025.6.24



CAMERON VALVES
Schlumberger Canada Limited
5003 - 93 Street
Edmonton Alberta Canada T6E 5S9
Phone 780.434.3476
www.slb.com/companies/cameron

CRN DESIGN REGISTRATION FOR GROVE VALVES

Scope:

GROVE B4D CRN 0C22914.2CL (exp. January 2033)
GROVE B5 CRN 0C22951.2CL (exp. February 2033)
GROVE B5R CRN 0C23020.2CL (exp. March 2033)
BODY CAVITY EQUALIZATION FITTING CRN 0H23278.2CL (exp. June 2033)

Manufacturing Location:

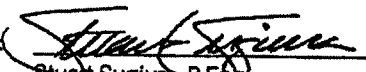
Edmonton, Canada (B4D, B5, B5R, EQUALIZATION FITTING)
Ville Platte, USA (B4D, B5, B5R)
Voghera, Italy (B4D, B5, B5R)
Welfang, China (B4D, B5, B5R)

The Cameron valve manufacturing facilities listed above are authorized to manufacture Grove B4D, B5, and B5R valves within the scope of CRN design registrations 0C22914.2CL, 0C22951.2CL, and 0C23020.2CL. These products are marked in accordance with API 6D and/or CSA Z245.15 as applicable and are pressure tested at the factory in accordance with those standards.

The Cameron valve manufacturing facility in Edmonton is also authorized to install a standard body cavity pressure equalization fitting within the scope of CRN design registration 0H23278.2CL. This fitting assembly is subject to pressure testing at ASME B16.34 test pressures and durations after installation on the valve.

It is the responsibility of the customer and the valve's end user to either specify or perform any additional testing that may be required by the code of installation for their application.

Approved by:


Stuart Sugiura, P.Eng.
Senior Valves Engineer

Date: 24 May 2024

05/23/2024

5A-EX-26 Rev 01

SLB-Private

Body: 122352167001

苏州恩威尔机械有限公司 Suzhou Envil Machinery CO., LTD		INSPECTION CERTIFICATE QUALITY CONTROL DEPARTMENT 检验证明书 (质量控制部)				Company's QMS certified body— ISO9001 No公司质量管理体系证书号:										
Cert. No. 证书号: 20250218006		Date 日期: 2025/2/18		CNBJ313986-U		表 码: QP010-09 版本号: 2023A										
Purchaser 买方: Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦龙石油机械有限公司		Purchase Order No. 订单号: 4514628084		Certificate type 证书类型: EN 10204		V		3.1 3.2 2.2								
		Our Job No. 工单号:		Melting practice 熔炼方法		V		LD-LF-RH-CCM								
CHEMICAL ANALYSIS 化学分析				Test Specification 测试标准 ASTM A751												
Material 材料规范: ASTM A350 GR LF2 CLASS 1		Heat No. 炉号: 12210568C H.T. Lot 热处理批次号: R250207T4		Raw Material Supplier 原材料供应商: JIANGSU SHAGANG GROUP HUAIGANG Special Steel Co., Ltd												
MS-021239-02-04 Rev.03	C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	H
Min	/	0.60	0.15	/	/	/	/	/	/	/	/	/	/	/	/	/
Max	0.23	1.35	0.30	0.40	0.30	0.02	0.025	/	0.40	0.12	/	/	0.03	0.02	/	/
Heat Analysis 浇包分析	0.18	0.99	0.24	0.01	0.03	0.006	0.008	/	0.030	0.004	/	/	0.006	0.0001	/	/
Product Analysis 产品分析	0.19	0.98	0.23	0.02	0.03	0.005	0.007	/	0.031	0.003	/	/	0.005	0.0001	/	/
碳当量 C.E. (Long) =	0.364		Cr+Mo=		0.033		Cu+Ni+Cr+Mo+V=		0.089							
HEAT TREATMENT 热处理																
Type 类型	Temperature (°C) 温度				Time (Hour) 时间				Cooling Method 冷却方法							
Quench	900 °C				4.0 H				Water Cooling							
Temper	610 °C				8.0 H				Air Cooling							
材料规范	抗拉强度 Tensile Strength (MPa)	屈服强度 Yield Strength (0.2% offset) (MPa)	伸长率 Elongation (%)	收缩率 Reduction of Area (%)	HARDNESS (HB) 硬 度			IMPACT TEST 冲击测试								
Min 最小	485	277	22	30	143			Dir. 方向	Type 类型	Temp °C 温度	Avr 平均	Min 最小	Actual 实测(J)			
Max 最大	655	/	/	/	197			L	V	-46 °C	27J	27J	156			
ACTUAL 实测	525	339	34	77	160 163 158								128			
Test Piece No. 试样号	Gauge length 标距(mm)	Diameter 直径(mm)	length of reduced section 缩颈 (mm)		Temp 温度(°C)		Direction 方向									
L250207003	50	12.48	60		room 室温		L									
From Piece 本体 HARDNESS (HB) 硬度	159HBW/P 10/3000	161HBW/P 10/3000	Remark: At least two pieces per batch when the batch is greater than one piece 当批次大于1件 时, 每批次至少检两件													
Analysis for Nickel plating 镀层分析																
Plating specification 电镀规范 (需要填)	Thickness 厚度 (/) in		Heat Treatment Temperature 热处理温度				Heat Treatment time 热处理时间									
/		/		/				/								
PO Item No. 订单行号	Forging ratio 锻造比	Description 描述				Part Number 部件号				Quantity 数量						
150	4.35	BODY B5 08" 900 A350 LF2				2604927-08-09-01 Rev.02				2						
Remarks 备注:																
Material according to the spec: NACE MR0175/ISO15156. ASTM A350 GR LF2 Class 1																
Material符合规范:																
Heat treatment furnaces calibrated according to API 6D 25th Ed. 热处理校验符合API 6D 25th.																
We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order, unless otherwise stated, material free from radioactive contamination.																
在此声明我们以上订单所订材料均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明。材料没有放射性污染。																
Surveyor's Name, Signature 检查员名字, 签字				Inspector's Name, Signature 审核员名字, 签字				Date 日期								
检查员签字日期: 2025/2/18				审核员签字日期: 2025/2/18				2025/2/18								



PL 78-11-17-5 7 FEB 1964

APPROVED
Gandamuthu . 06-04-2025, 16:54:29

Ball: 122352167001

 球豹阀门有限公司 QIUBAO VALVE CO., LTD.		INSPECTION CERTIFICATE QUALITY CONTROL DEPARTMENT 检验证明书 (质量控制部)		Company's QMS certified body— ISO9001 No公司质量管理体系证书号: 253697-2018-AQ-RQC-RvA												
Cert. No. 证书号: 2025-02570		Date 日期: 2025-04-30														
Purchaser 买方: Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦龙石油机械有限公司		Purchase Order No. 订单号: 4514628071		Certificate type 证书类型: EN 10204 ☐ 2.2 ☒ 3.1 ☐ 3.2												
		Our Job No. 工单号:		Melting practice 熔炼方法 LD-LF												
CHEMICAL ANALYSIS 化学分析				Test Specification 测试标准 ASTM A751												
Material MS-021239 REV.03AMS-021239-02-34 REV.03AMS-021000-01 REV.16 材料规范: ASTM A350 LF2 Class1-24b				Heat No. 炉号: 32106085 H.T.Lot 热处理批次号: 25021503												
Raw Material Supplier 原材料供应商: 新冶特钢有限责任公司																
	C	Mn	Si	S	P	Cr	Ni	Mo	Cu	V	Nb	N	Ti	Al	Sn	H
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm
MIN		0.60	0.15													
MAX	0.230	1.35	0.30	0.020	0.025	0.30	0.40	0.12	0.40	0.030	0.020					
Heat Analysis 逐包分析	0.19	0.95	0.21	0.005	0.015	0.93	0.01	0.01	0.03	0.003	0.001					
Product Analysis 产品分析	0.185	0.986	0.237	0.0032	0.0111	0.0364	0.0239	0.0163	0.0244	0.0045	0.0026					
碳当量 C.E. (Long) = 0.364				Cr+Mo = 0.0527				Cu+Ni+Cr+Mo+V = 0.1055								
HEAT TREATMENT 热处理																
Type 类型		Temperature (°C) 温度				Time (Hour) 时间				Cooling Method 冷却方法						
Quenching		890				3				Water						
Tempering		600				3.5				Air						
MECHANICAL TEST 机械性能测试				Test Specification 测试标准 ASTM A370				From QTC 试棒: ☒								
Material 材料规范	抗拉强度 Tensile Strength (MPa)	屈服强度 Yield Strength (0.2% offset) (MPa)	伸长率 Elongation (%)	收缩率 Reduction of Area (%)	HARDNESS (HB) 硬度				IMPACT TEST 冲击测试							
									Dir. 方向	Type 类型	Temp °C 温度	Avg 平均	Min 最小	Actual 实测 (J)		
Min 最小	485	277	22	30	143				L	V	-46	27	27	126		
Max 最大	655				197									144		
ACTUAL 实测	548	356	33.5	72	160	163	161	150								
Test Piece No. 试样号		Gauge length 标距 (mm)		Diameter 直径 (mm)		length of reduced section 缩颈 (mm)		Temp 温度 (°C)		Direction 方向						
32106085/25021503		50		12.5		60		room 室温		L						
From Product 本体 HARDNESS (HB) 硬度		161		164		Remark: At least two pieces per batch when the batch is greater than one piece 当批次大于1件时, 每批次至少检两件										
Analysis for Nickel plating 镀层分析																
Plating specification 电镀规范 (SP-054677 REV.06&SP-054677-01-02-75 REV.03)		Thickness 厚度 (0.0030) in		Heat Treatment Temperature 热处理温度 (°C)				Heat Treatment time 热处理时间 (h)								
实测值 Actual		0.0031-0.0033 in		300 °C				1 hour								
PO Item No. 订单行号		Forging ratio 锻造比		Description 描述				Part Number 零件号				Quantity 数量				
00100		4.56:1		BALL B5 08" 900 A350 LF2 3MIL				2604926-08-09-03(03)				2				
Remarks 备注: Material according to the spec: NACE MR0175/ISO15156 ASTM A350 Grade LF2 CLASS 1 API 20B FSL-4 材料符合规范: Heat treatment furnaces calibrated according to API 6D 25th Ed. Annex H and API 6A 21st Ed. Annex M 热处理校核符合 API 6D 25th 版附录 H 和 API 6A 21st 附录 M. We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise stated. Material is free from radioactive contamination 在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 材料没有放射性污染。 Surveyor's Name, Signature & Date 检查员签字日期: Li Shijing 2025-04-30 Inspector's Name, Signature & Date 审核员签字日期: Zhu Lushan 2025-04-30																

APPROVED
Gandamulhu, 14-07-2025, 13:16:46



QIUBAO VALVE CO.,LTD.

球豹阀门有限公司

ENP INSPECTION REPORT

化学镀镍磷检测报告

客户名称 Customer Name	Shandong Cameron Petroleum Equipment Pte Ltd	产品类型 Production Name	Ball of Ball Valve	编号Report No.	ENP-A209564
订单号\行号 PO. No.\Item No.	4514628071\00100	材料 Production Material	LF2 Class1 ENP0.003in	日期 Inspection Date	2025/4/30
型号规格 Production Size	B5 08" 900	数量 Production Quantity	2PCS	镀后热处理 Post Heat Treatment	300℃/1hour
零件号 Part No.	2604926-08-09-03(03)	电镀标准 Carry out Standard	SP-054677 REV.06&SP- 054677-01-02-75 REV.03	炉号Heat No.	32106085
				批号Batch No.	JN209564

1.附着力测试 Adhesion Test

按照ASTM B571标准在试样上进行,将试样弯曲180°后,未发现镀层有剥落现象。

As per ASTM B571 the test strip was bent approximately 180 degree over a mandrel. None peeling or flaking of the deposit was observed.

2.孔隙率测试 Porosity Test

根据ASTM B733 9.6.1标准进行孔隙试验,未发现明显的蓝色斑点或条纹。

Ferroxyl test was carried out as per ASTM B733 9.6.1, none of sharply defined blue spots or streaks.

3.外观检测 Visual Inspection

涂层光滑,附着,无可见的水疱,凹坑,结节,孔隙,裂缝。无明显的搭缝、接缝、裂缝或破裂。

Coating is smooth,adherent and free from visible blisters,pits,nodules,porosith ,cracks ,uncoated areas and other detects .
No visible laps, seams, cracks or bursts.

4.厚度检测 Thickness Measurement

用测厚仪在同炉试样上进行检测,经检测厚度为0.0031-0.0033in。

The thickness was measured on test strip by thickness testing machine. It was 0.0031-0.0033in

5.成分分析 Composition Analysis

成份检测在同炉试样中进行,经检测磷的含量约为10.89%,其余为镍含量。

The chemical composition for the deposit is 10.89% phosphorous and balance is nickel.

6.硬度检测 Hardness Measurement

电镀后每100克硬度为750-800HV。

Hardness after plating average 750-800HV at 100g load.

7.试片在室温下浓度为70%的硝酸溶液中浸泡30秒。未发现试片表面有变黑等现象。

The test strip immersed in 70% nitric acid solution at ambient temperature for 30 seconds ,and didn't turn black .

结论Conclusion 合格(√)Conformity 不合格()Non-Conformity

检验员
Inspector

Cao Jike

质保经理
Q.A Manager



QB/QR18.21 Rev.B/0

Stem: 122352167001

供应商名称 Supplier Name 南京青特复合材料股份有限公司 Nanjing CompTECH Composites Corporation		检验证明书-质量控制部 Inspection Certificate-Quality Control Dept. 证书号 / Cert. No. 10008FG250080-6 日期 / Date 2025/4/23				ISO9001公司质量管理体系证书号 Company's QMS Certificate No. CN17/21222.00													
买方 Purchaser 山东卡麦龙石油机械有限公司 Shandong Cameron Petroleum Equipment Pte Ltd		订单号 Purchase Order No. 4514628077				证书类型 Certificate Type: EN 10204 V 3.1 3.2 2.2													
工单号 Our Job No. 10008FG250080-6		熔炼方法 Melting Practice V EF																	
化学分析 Chemical Analysis		测试标准 Test Specification: ASTM A751																	
材料规范 MS Code Material Specification MS-021272-01(03)		炉号 Heat No. 3301711 热处理批次号 H.T.Lot No. 240324H04				原材料供应商 Raw Material Supplier Shandong Shouguang Juneng													
		C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	B	V	Nb	N	Nb+Ta		
最小 Min.		0.38	0.50			0.90					0.15								
最大 Max.		0.45	0.90	0.40	0.50	1.20	0.035	0.035		0.35	0.25			0.05					
浇包分析 Heat Analysis		0.42	0.89	0.25	0.03	0.99	0.007	0.011		0.04	0.200			0.007					
产品分析 Product		0.43	0.88	0.26	0.01	0.97	0.009	0.013		0.02	0.21			0.007					
碳当量 CE (Long)=		Max.:				Cr+Mo=				Max.:				Cu+Ni+Cr+Mo+V=				Max.:	
热处理 Heat Treatment																			
类型 Type 淬火 (Quenching) 回火 (Tempering)		温度 Temperature (°C) 880 680				保温时间 Keep Time (Hour) 2 3				冷却方法 Cooling Method 水冷 (Water Cooling) 空冷 (Air Cooling)									
机械性能测试 Mechanical Test																			
材料规范 MS Code MS-021272-01(03)		抗拉强度 Tensile Strength Mpa		屈服强度 Yield Strength 0.2% offset Mpa		伸长率 Elongation (%)		收缩率 Reduction of Area (%)		硬度 Hardness HBW/P 10/3000		冲击测试 Impact Test							
		最小 Min.		677		520		15		45		HBW		237		L V -46 27 27			
		最大 Max.														128			
		实测 Actual		705		546		33		71		224 228 227				108 118			
试样号 Test Piece No. 3301711 240324H04		标距 mm Gauge Length 50		直径 mm Diameter 12.5		缩颈 mm Length Of Reduced Section 60		温度 °C Temp 室温 Room Temp		方向 Direction L									
本体硬度 Hardness From Piece HBW/P 10/3000		270		228		Remark: At Least Two Pieces Per Batch When The Batch Is Greater Than One Piece. 当批次大于1件时, 每批次至少检2件.													
镀层分析 Analysis For Nickel Plating																			
电镀规范 (需要填) Plating Specification		要求厚度 Inch Required Thickness		实测厚度 Inch Actual Thickness		热处理温度 °C Heat Treatment Temperature		热处理时间 小时 Heat Treatment Time Hour											
SP-054677-01-01-75(03)		0.003		0.003															
订单行号 PO Item No.		锻造比 Forging Ratio		描述 Description				部件号 Part Number				数量 Quantity							
120		4:1		STEM B5 75 MM 42CrMo4 3 MIL ENP				2604929-75-07-02(02)				2							
备注 Remarks:																			
材料符合性 Material According To The Spec		NACE MR0175/ISO15156				42CrMo4 UNI EN 10083-1													
热处理校准符合 API 6D 25th Ed. Heat Treatment Furnaces Calibrated According To API 6D 25th Ed.																			
在此声明我们以上订单所描述的货品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明, 材料没有被放射性污染。 We hereby certify that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise stated, material free from radioactive contamination.																			
检查员签字/日期 Surveyor's Name, Signature & Date Zaiqin Chen (需要) 2025/4/23									审核员签字/日期 Inspector's Name, Signature & Date Weiming Sun 2025/4/23										





聚恩辈

温州聚恩辈实业有限公司检测报告

JUENBEI INDUSTRIAL

WENZHOU JUENBEI INDUSTRIAL CO.,LTD. ENP INSPECTION REPORT



No. MS243801 RO

客户名称 Customer Name	南京肯特复合材料 股份有限公司	产品类型 Production Name	STEM	报告编号 Report No.	02503003709-08
订单号\行号 P.O. No.\Item No.	4514628077/00120	材料+电镀厚度 Material+ENP Thickness	42CrMo4 UNI EN 10083-1+0.075	日期 Inspection Date	2025-3-10
型号规格 Production Size	2604929-75-07-02	数量 Production Quantity	2 PCS	热处理温度/时间 Heat/Time	N/A
硬度要求 Hardness Require	≥500HV	电镀标准 Carry out Standard	SP-054677-01-01-75 03	电镀批号 Batch No.	02503003795
检测记录 Inspection Record					
1. 附着力测试 Adhesion Test					
附着力测试根据ASTM B571标准在试样上进行,将试样弯曲180°后,未发现镀层有剥落现象。 Adhesion test as per ASTM B571 the sample was bend at 180 degree over a Mandrel of Same size as sample No peeling was observed.					
2. 孔隙测试 Porosity Test					
根据ASTM B733 9.6.1标准进行孔隙试验,未发现蓝色斑点。 Ferroxyl test carried out as per ASTM B733 standard to check for porosity None were found.					
3. 外观检测 Visual Test					
外观检测涂层光滑、粘附,无可见的水泡、凹坑、结节、气孔、裂缝、未涂层区域和其他缺陷。 The appearance inspection shows that the coating is smooth and adherent, without visible blisters, pits, nodules, pores, cracks, uncoated areas and other defects.					
4. 厚度检测 Thickness Test					
用测厚仪在同炉实物上进行检测,经检测厚度为0.076-0.079mm The thickness was measured on matter and was found to be 0.076-0.079mm With Thickness Testing Machine.					
5. 成份检测 Composition Test					
成份检测在同炉试样中进行,经检测磷的含量约为11.09%,其余为镍含量。 The job were plated with E.N.P having average phosphorus content of 11.09% and Remaining being nickel.					
6. 硬度检测 Hardness Test					
产品电镀后检测其平均硬度值为527HV100 Hardness after plating average 527HV at 100 gm load.					
7. 硝酸试验 NITRIC ACID TEST					
样板浸入硝酸中30秒,未发现发黑。 The sample plate is immersed in nitric acid for 30 seconds, No blackening found.					
结论 Conclusion	合格 (✓) Conformity 不合格 () Non-Conformity				
检验员 Inspector:	2025年	审 批 Approved by:	袁文亮		



QR-QEO02-18-01 Rev.A/0

~~PL IV/II-17-67 RSV B/O~~

Page 1 of 1



青田永拓金属表面技术处理有限公司
Qingtian Yongtuo Metal Surface Technical Treatment Co., Ltd.
化学镀镍磷检测报告 ENP INSPECTION REPORT

日期 Date: 2025-05-06

客户名称 Customer Name	江苏应流机械制造有限公司 限责任公司	产品类型 Production Name	阀座组件 SEAT ASSY	报告编号 Report No.	250506-11
订单号\行号 P.O. No. \Item No.	4514628082/510	材料+电镀厚度 Material+ENP Thickness	A350 LF2 CL1 +ENP0.075+ERTALON 6XALmm	跟踪卡号 Route NO	/
型号规格 Production Size	08-0900 9G-P22D08909	数量 Production Quantity	4件	电镀批号 Batch No	YT14148
硬度要求 Hardness Require	≥500	电镀标准 Carry out Standard	SP-054677-01-01- 75 (Rev06/03)	后热处理温度/时间 Post Heat Treatment.	N/A
检测记录 Inspection Record					
1. 附着力测试 Adhesion Test 附着力测试根据ASTM B571标准在试样上进行, 将试样弯曲180°后, 未发现镀层有剥落现象。 Adhesion test as per ASTM B571 the sample was bend at 180 degree over a Mandrel of Same size as sample No peeling was observed.					
2. 孔隙测试 Porosity Test 根据ASTM B733 9.6.1标准进行孔隙试验, 未发现蓝色斑点。 Ferroxyl test carried out as per ASTM B733 standard to check for porosity None were found.					
3. 外观检测 Visual Test 外观检测发现表面光滑并粘附, 未发现表面存在水泡、凹坑、结节、气孔、裂纹等缺陷。 Appearance inspection found that the surface was smooth and adhered, and no surface defects such as blisters, pits, nodules, pores and cracks were found.					
4. 厚度检测 Thickness Test 用测厚仪在同炉实物上进行检测, 经检测厚度为0.075-0.080mm The thickness was measured on matter and was found to be 0.075-0.080mm With Thickness Testing Machine.					
5. 成份检测 Composition Test 成份检测在同炉试样中进行, 经检测磷含量约为10.54%, 镍的含量约为89.46%。 The content of phosphorus and nickel is about 10.54% and 89.46% respectively.					
6. 硬度检测 Hardness Test 产品电镀后检测其平均硬度值为:550HV100。 Hardness after plating average :550HVat 100 gm load					
7. 硝酸实验检测 Nitric acid test test 将镀有75微米化学镀镍的试样条侵入70%的硝酸溶液中静放30秒, 取出后, 表面未发现变色。 The sample plated with 75 micron chemical nickel plating was inserted into the 70% Nitric acid solution for 30 seconds. After taking out, no color difference was found on the surface.					
结论 Conclusion	合格(√) Conformity 不合格() Non-Conformity				
检验员: Inspector	项余梅	审批: Approved by	练徐波	 单位盖章 Stamp of organization	

YT-QES-R/P014-05Rev.A/1

WL 18/11-17-67 REV. 3/0

APPROVED
GAndamuthu, 27-04-2025, 10:25:11



Shandong Cameron Petroleum Equipment Pte Ltd
No. 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0601

CERTIFICATION OF COMPLIANCE API 6D 25TH EDITION

Customer Name: STAFFORD - 13127 TRINITY DR
CC STO No./PO No. 4710753518
Sales Order Reference Number: N#A
Item Number: 40
Work order Number: 122432354
Quantity of Valves: 2

We hereby certify that the materials for component parts and the manufacture of said valves are in compliance with Cameron valves specifications and drawings and meet all requirements of the above referenced customer order.

We further certify that the above listed valve(s) are in full conformance with American Petroleum Institute (API) 6D 25th Edition and all addenda (as of this date hereby certified), have been API monogrammed and has/have successfully passed all applicable test in accordance with requirements thereof, including additional customer requirements agreed upon in the above indicated order.

ID	Serial Number	Tag No.	Valve Part Number	Valve Type	Valve Size	Pressure Class	End Conn.	Trim	Code of pressure-temperature rating	Balance hole	Design code of pressure boundary bolting	Design code of pressure containing parts
1	122432354001	N/A	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim	ASME B16.34	Yes	EN12516-2	ASME VIII Div 1
2	122432354002	N/A	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim	ASME B16.34	Yes	EN12516-2	ASME VIII Div 1

Signature:

Name: Song Jian

Date: 2026.1.8

	EU Declaration of Conformity Pressure Equipment Directive 2014/68/EU		Doc.No.: SLB-QF-A344-093-1	
			Revision: 14	
	Product Category III Conformity Assessment Module H, used for group I.		Customer Name:	
			STAFFORD - 13127 TRINITY DR	
		PO number: 4710753518		
		Sales Order Number: N#A		
Notified Body	Manufacturer			
LRQA Nederland B.V. at George Hintzenweg 77, 3068 AX Rotterdam, The Netherlands. Identification No.0343 Certification No. 0343/PED/0038/PED/MUM/071 0044/1	Shangdong Cameron Petroleum Equipment PTE Ltd. No. 2577 Gaoxin 5th Road, Gaoxin District. Weifang City, Shandong Province, P.R. China,261041 Tel:86-536-2080601 Tax:86- 536-2080617			
We hereby declare that product(s) described below are in conformance with the pressure Equipment Directive 2014/68/EU This declaration of conformity is issued under the sole responsibility of the manufacturer				
Product Model Number	Qty	Serial Number	Technical File No.	Rev.
Grove B5	1	122432354001	B5 Technical file	1
Grove B5	1	122432354002	B5 Technical file	1
Other EC Directives: _____ Ref. Design Standards: ASME Section VIII Div I, Section II Part D:API 6D, Valves <u>Asme16.34, Valve-Flanged, Threaded, Welding Ends.</u>				
QA Representative: <u>Song Jian</u>		Title: <u>QA Manager</u>		
Signature: 		Date: <u>2026.1.8</u>		



Shandong Cameron Petroleum Equipment Pte Ltd
No. 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0601

CERTIFICATION OF COMPLIANCE NACE MR0175/ISO15156

Customer Name:
CC STO No./PO No.
Sales Order Reference Number:
Work order number:

STAFFORD - 13127 TRINITY DR
4710753518
N#A
122432354

Cameron certifies the subject product in compliance with pre-qualified materials of NACE MR0175/ISO15156. This certification does not imply or warrant the application of the product in compliance with NACE MR0175/ISO15156 service conditions in which the customer/user installs the product.

ID	Serial Number	Valve Part Number	Valve Type	Valve Size	Pressure Class	End Conn.	Trim
1	122432354001	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim
2	122432354002	2510873-08-09-03	Ball, Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim

Signature: _____

Name: Song Jian

Date: 2026.1.8



Shandong Cameron Petroleum Equipment Pte Ltd
No. 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0601

LETTER OF COMPLIANCE For: GROVE BRAND TRUNNION VALVES Model B4, B5,B7-Sour Low Temperature Service

Customer Name:
CC STO No./PO No.
Sales Order Reference Number:
Item Number:
Work order number:

STAFFORD - 13127 TRINITY DR
4710753518
N#A
40
122432354

Cameron Valves & Process Systems hereby certifies that the design, manufacture, sampling, testing and quality control inspection of the GROVE brand CSA Z245.15-22 ball valves for Sour Low Temperature Service meet the requirements of CSA Z245.15 Sour Service, VPS Engineering Standards, and customer purchase order requirements. As part of compliance to these specifications, the valves are marked in accordance with MSS SP-25. As part of conformance to these specifications, the valves are marked in accordance with the requirements of API 6D and CSA Z245.15:22 Table 1 and are suitable for installation in CSA Z662 pipeline systems."

The materials of construction for these valves conform to the metallurgical requirements of NACE MR0175/ISO1516 (Parts 1 & 2 for Carbon & Alloy Steels in SSC Region 3 Sour Service and Parts 1 & 3 for CRA's as specified in CSA Z245.15 Chapter 10). According to NACE MR0175/ISO 15156, it is the customer's / user's responsibility to ensure that the valve materials will be satisfactory in the intended service.

The minimum service temperature of these valves is -46°C/50°F. Impact testing has been performed in accordance with applicable ASTM requirements and those of CSA Z245.15 for Category II -45°C Service.

The ASTM A320 Grade L7M studs and cap screws and ASTM Grade A194 7ML nuts used in the products specified above have been Quality Control inspected and meet the requirements of the applicable ASTM specifications-as is already implied fully by the statements above regarding both CSA and NACE standards compliance.

ID	Serial Number	Valve Part Number	Valve Type	Valve Size	Pressure Class	End Conn.	Trim
1	122432354001	2510873-08-09-03	Ball,Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim
2	122432354002	2510873-08-09-03	Ball,Trunnion	8"X8"	900	RF x RF	G02-CS Body 3 MIL ENP Trim

Approved by: _____

Song Jian
Quality manager

Date: 2026.1.8 _____

Body: 122432354002

苏州恩威尔机械有限公司 Suzhou Envil Machinery CO.,LTD		INSPECTION CERTIFICATE QUALITY CONTROL DEPARTMENT 检验证明书 (质量控制部)						Company's QMS certified body— ISO9001 No公司质量管理体系证书号:															
Cert. No. 证书号:20250909004		Date 日期: 2025/9/9						CNBJ313986-U 表 码: QP010-09 版本号: 2023A															
Purchaser 买方: Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦龙石油机械有限公司		Purchase Order No. 订单号 : 4514888338						Certificate type 证书类型: EN 10204															
		Our Job No. 工单号:						Melting practice 熔炼方法															
		V						LD-LF-VD															
CHEMICAL ANALYSIS 化学分析												Test Specification测试标准 ASTM A751											
Material 材料规范:ASTM A350 GR LF2 CLASS 1				Heat No.炉号:52305511 H.T.Lot热处理批次号:R250904T3				Raw Material Supplier 原材料供应商:XIN YE Special Steel Co.,Ltd															
MS-021239-02-04 Rev.03		C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	H						
		%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%						
Min		/	0.60	0.15	/	/	/	/	/	/	/	/	/	/	/	/	/						
Max		0.23	1.35	0.30	0.40	0.30	0.02	0.025	/	0.40	0.12	/	/	0.03	0.02	/	/						
Heat Analysis 浇包分析		0.19	0.94	0.23	0.02	0.03	0.009	0.014	/	0.030	0.01	/	/	0.003	0.002	/	/						
Product Analysis 产品分析		0.18	0.95	0.22	0.03	0.040	0.007	0.013	/	0.020	0.011	/	/	0.002	0.002	/	/						
碳当量C.E. (Long) =		0.352		Cr+Mo=		0.051		Cu+Ni+Cr+Mo+V=				0.103											
HEAT TREATMENT 热处理																							
Type 类型		Temperature (°C) 温度						Time (Hour) 时间						Cooling Method 冷却方法									
Quench		920 °C						4.0 H						Water Cooling									
Temper		610 °C						8.0 H						Air Cooling									
材料规范		抗拉强度 Tensile Strength (MPa)		屈服强度Yield Strength (0.2% offset) (MPa)		延长率 Elongation (%)		收缩率 Reduction of Area(%)		HARDNESS (HB) 硬 度			IMPACT TEST 冲击测试										
													Dir. 方向	Type 类型	Temp °C 温度	Avr 平均	Min 最小	Actual 实测(J)					
Min最小		485		277		22		30		143			L	V	-46 °c	27J	27J	120					
Max最大		655		/		/		/		197								130					
ACTUAL实测		510		320		31		74		158 163 160								100					
Test Piece No. 试样号		Gauge length 标距(mm)				Diameter 直径(mm)		length of reduced section 缩颈 (mm)				Temp 温度(°C)				Direction方向							
L25090401		50				12.46		60				room 室温				L							
From Piece本体 HARDNESS (HB) 硬度		158HBW/P 10/3000		163HBW/P 10/3000		Remark: At least two pieces per batch when the batch is greater than one piece当批次大于1件 时, 每批次至少检两件																	
Analysis for Nickel plating镀层分析																							
Plating specification 电镀规范 (需要填)				Thickness厚度 (/) in				Heat Treatment Temperature 热处理温度				Heat Treatment time 热处理时间											
/				/				/				/											
PO Item No. 订单 行号		Forging ratio 锻造比		Description 描述						Part Number 部件号				Quantity 数量									
30		4.27		BODY B5 08" 900 A350 LF2						2604927-06-09-01 Rev.02				5									
Remarks备注: Material according to the spec: NACE MR0175/ISO15156, ASTM A350 GR LF2 CLASS 1 材料符合规范: Heat treatment furnaces calibrated according to API 6D 25th Ed. 热处理校验符合API 6D 25th. We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise stated.material free from radioactive contamination. 在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明。材料没有被放射性污染。 Surveyor's Name, Signature & Date 检查员签字日期: 周玲玲 2025/9/9 Inspector's Name, Signature & Date 审核员签字日期: 2025/9/9																							

APPROVED
GAndamuthu , 17-12-2025, 12:26:45

JIANGSU YINGLIU MACHINE MANUFACTURING CO., LTD. 江苏应流机械制造有限公司				INSPECTION CERTIFICATE QUALITY CONTROL DEPARTMENT 检验证明书 (质量控制部)				Company's QMS certified body—ISO9001 No公司质量管理体系证书号: No29333-2008-AO-RGC-RvA/Rev.1													
Purchaser: 买方 Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦龙石油机械有限公司				Cert. No. 证书号:	25090521		Date 日期:	2025. 09. 08		Material Specification 材料规格:		MS-021239-02-04 REV. 03									
				Purchase Order No. 订单号:		4514868938				Certificate type 证书类型:		EN 10204 □2.2. ■3.1 □3.2									
				Our Job No. 工单号:		YC2780300				Melting practice 熔炼方法:		■E.F. □E.F.V.D.									
CHEMICAL ANALYSIS 化学分析										Test Specification 测试标准: ASTM A751											
Material 材料规格:		ASTM A350 GRADE LF2 CLASS 1		Heat No. 炉号:		250602283		H.T.Lot 热处理批次号:		250903-A2C2			Raw Material Supplier 原材料供应商:			BEI MAN					
MS-021239-02-04 REV.03		C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	H				
		%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm			
Min			0.57	0.13																	
Max		0.23	1.39	0.32	0.43	0.33	0.02	0.025		0.43	0.13			0.03	0.02						
Heat Analysis 浇包分析		0.190	0.960	0.250	0.0400	0.100	0.002	0.0140		0.020	0.0100			0.0100	0.0010						
Product Analysis 产品分析		0.192	0.960	0.236	0.0325	0.1480	0.0025	0.0120		0.0063	0.0019			0.00200	0.0058						
碳当量C.E. (Long) (≤0.43)		0.385				Cr+Mo (≤0.32)		0.150		Cu+Ni+Cr+Mo+V (≤1.00)			0.191			PREN		-			
HEAT TREATMENT 热处理 Heat treatment equipment comply with API 6D 25 edition and API 6A 21 edition.																					
Type 类型		Temperature (°C) 温度							Time (Hour) 时间					Cooling Method 冷却方法							
Quenching		895							1.83h					Water cooled							
Tempering		625							2.70h					Air cooled							
MECHANICAL TEST 机械性能测试		Test Specification 测试标准: ASTM A370-24a							From QTC 试样: ■ (100mm*100mm*200mm)					From Piece 本体: □							
Location 位置/方向		UTS σ b 抗拉强度		Ys σ s (0.2% offset) 屈服强度		EL δ 5 伸长率		ROA ϕ 收缩率		IMPACT TEST 冲击测试 10x10x55 mm					HARDNESS (HBW) 硬度						
1/2 T longitudinal		(Mpa)		(Mpa)		(%)		(%)		Location 位置/方向		Type 类型		Temp °C 温度		Min 最小		Avr 平均	Min 最小	Max 最大	
Min		485		277		22		30		1/2 T longitudinal		V		-46°C		27		27	143	197	
Max		655		/		/		/													
ACTUAL 实测		UTS σ b 抗拉强度		Ys σ s (0.2% offset) 屈服强度		EL δ 5 伸长率		ROA ϕ 收缩率		IMPACT TEST 冲击测试 10x10x55 mm					HARDNESS (HBW) 硬度						
		(Mpa)		(Mpa)		(%)		(%)		Location 位置/方向		Type 类型		Temp °C 温度		Joules 焦耳					
		553		386		35		74		1/2 T longitudinal		V		-46°C		134 127 118 126		160/158/157			
Yield/Tensile strength ratio 屈服抗拉强度比				-				Lateral expansion(mm) 侧膨胀				-									
Test Piece No. 试样号		Gauge length 标距 (mm)		Diameter 直径 (mm)		Length of reduced section 缩颈 (mm)		Temp 温度 (°C)		Direction 方向											
9-13#		50.00		12.52		60		20		longitudinal 纵向											
From Piece 本体 HARDNESS 硬度 (HBW/P 10/3000)		160		158		Remark: At least two pieces per batch when the batch is greater than one piece 当批次大于1件时, 每批次至少检两件															
Analysis for Plating 镀层分析																					
Plating specification 电镀规范		Thickness 厚度 (in)		Heat Treatment Temperature 热处理温度 (°C)		Heat Treatment time 热处理时间 (h)		Surface Hardness 产品表面硬度 (HV100)													
/		Standard 标准		/		/		/		Min 最小		/		Max 最大		/					
		Actual 实测		/		/		/		/		/		/		/					
PO Item No. 订单行号		Your Item No. 你们的行号				Description 描述				Forging ratio 锻造比		Part Number 部件号		Quantity 数量							
150						CLOSURE B5R 06" 900 RF A350 LF2				4.4:1		2463329-06-09-01		12							
250						CLOSURE B5R 08" 900 RF A350 LF2				4.2:1		2463329-08-09-01		4							
Remarks 备注:																					
Material according to the spec: 材料符合规范:				ASTM A350 GRADE LF2 CLASS 1																	
Heat treatment furnaces calibrated according to API 6D 25th Ed. Annex F and API 6A 21th Ed. Annex M 热处理炉符合 API 6D 25th 版附录 F 和 API 6A 21th 附录 M.																					
We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise stated. Material free from radioactive contamination 在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明, 材料没有被放射性污染。																					
Prepared by 编制签字:		李萍				Proofread 校对签字:				赵海				Inspector's Name, Signature 审核员签字:		张					

Ball: 122432354002

APPROVED
SMukundan2, 05-12-2025, 17:08:13

 球豹阀门有限公司 QIUBAO VALVE CO., LTD.		INSPECTION CERTIFICATE QUALITY CONTROL DEPARTMENT 检验证明书 (质量控制部)		Company's QMS certified body— ISO9001 No公司质量管理体系证书号: 259697-2018-AQ-RGC-RvA												
Cert. No. 证书号:2025-07989		Date 日期:2025-11-19														
Purchaser 买方: Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦龙石油机械有限公司		Purchase Order No. 订单号:4514912198		Certificate type 证书类型: EN 10204 ☐ 2.2 ☒ 3.1 ☐ 3.2												
		Our Job No. 工单号:		Melting practice 熔炼方法 LD+LF+VD												
CHEMICAL ANALYSIS 化学分析				Test Specification 测试标准 ASTM A751												
Material: MS-021239 REV.03&MS-021239-02-04 REV.03&MS-021000-01 REV.16 材料规范: ASTM A350 LF2 Class1-24b				Heat No.炉号: 32109833A H.T.Lot热处理批次号:25100401												
				Raw Material Supplier 原材料供应商: 新冶特钢有限责任公司												
	C	Mn	Si	S	P	Cr	Ni	Mo	Cu	V	Nb	N	Ti	Al	Sn	H
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm
MIN		0.60	0.15													
MAX	0.230	1.35	0.30	0.020	0.025	0.30	0.40	0.12	0.40	0.030	0.020					
Heat Analysis 浇包分析	0.19	1.19	0.23	0.007	0.015	0.04	0.02	0.01	0.05	0.003	0.004					
Product Analysis 产品分析	0.187	1.18	0.217	0.0086	0.0125	0.0503	0.0238	0.006	0.0526	0.0062	0.0046					
碳当量C. E. (Long) =				Cr+Mo=				Cu+Ni+Cr+Mo+V=								
HEAT TREATMENT 热处理																
Type 类型		Temperature (°C) 温度				Time (Hour) 时间				Cooling Method 冷却方法						
Quenching		890				2.5				Water						
Tempering		620				3				Air						
MECHANICAL TEST 机械性能测试				Test Specification 测试标准 ASTM A370				From QTC试棒: ☒								
Material 材料规范	抗拉强度Tensile Strength (MPa)	屈服强度Yield Strength (0.2% offset) (MPa)	伸长率 Elongation (%)	收缩率 Reduction of Area(%)	HARDNESS (HB) 硬度			IMPACT TEST 冲击测试								
								Dir. 方向	Type 类型	Temp°C 温度	Avr 平均	Min 最小	Actual 实测(J)			
Min最小	485	277	22	30	143			L	V	-46	27	27	130			
Max 最大	655				197								144			
ACTUAL实测	515	330	30.5	68	155	154	151						148			
Test Piece No. 试样号		Gauge length 标距(mm)		Diameter 直径(mm)		length of reduced section 缩颈(mm)		Temp 温度(°C)		Direction 方向						
32109833A/25100401		50		12.5		60		room 室温		L						
From Product 本体 HARDNESS (HB) 硬度		156		155		Remark: At least two pieces per batch when the batch is greater than one piece当批次大于1件时, 每批次至少检两件										
Analysis for Nickel plating镀层分析																
Plating specification 电镀规范 (SP-054677 REV.06&SP-054677- 01-02-75 REV.03)		Thickness厚度 (0.0030) in			Heat Treatment Temperature 热处理温度(°C)				Heat Treatment time 热处理时间(h)							
实测值Actual		0.0031-0.0033in			300℃				1hour							
PO Item No. 订单 行号		Forging ratio 锻造比		Description 描述				Part Number 部件号				Quantity 数量				
00230		4.56:1		BALL B5 08" 900 A350 LF2 3MIL				2604926-08-09-03(03)				2				
Remarks备注: Material according to the spec: NACE MR0175/ISO15156 ASTM A350 Grade LF2 CLASS 1 API 20B FSL-4 材料符合规范: Heat treatment furnaces calibrated according to API 6D 25th Ed. Annex H and API 6A 21st Ed. Annex M 热处理校验符合API 6D 25th版附录H和API 6A 21st 附录M。 We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise stated. Material is free from radioactive contamination 在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明。材料没被放射性污染。																
Surveyor's Name, Signature & Date 检查员签字日期: Li Shijing 2025-11-19								Inspector's Name, Signature & Date 审核员签字日期: Zhi Liang 2025-11-19								



QIUBAO VALVE CO.,LTD.

球豹阀门有限公司

ENP INSPECTION REPORT

化学镀镍磷检测报告

客户名称 Customer Name	Shandong Cameron Petroleum Equipment Pte Ltd	产品类型 Production Name	Ball of Ball Valve	编号Report No.	ENP-A214759
订单号\行号 PO. No.\Item No.	4514912198\00230	材料 Production Material	LF2 Class1ENP0.003in	日期 Inspection Date	2025/11/14
型号规格 Production Size	B5 08" 900	数量 Production Quantity	2PCS	镀后热处理 Post Heat Treatment	300°C/1hour
零件号 Part No.	2604926-08-09-03(03)	电镀标准 Carry out Standard	SP-054677 REV.06&SP- 054677-01-02-75 REV.03	炉号Heat No. 批号Batch No.	32109833A YT15016

1.附着力测试 Adhesion Test

按照ASTM B571标准在试样上进行,将试样弯曲180°后,未发现镀层有剥落现象。

As per ASTM B571 the test strip was bent approximately 180 degree over a mandrel. None peeling or flaking of the deposit was observed.

2.孔隙率测试 Porosity Test

根据ASTM B733 9.6.1标准进行孔隙试验,未发现明显的蓝色斑点或条痕。

Ferroxyl test was carried out as per ASTM B733 9.6.1, none of sharply defined blue spots or streaks.

3.外观检测 Visual Inspection

涂层光滑,附着,无可见的水疱,凹坑,结节,孔隙,裂缝。无明显的搭缝、接缝、裂缝或破裂。

Coating is smooth,adherent and free from visible blisters ,pits,nodules,porosith ,cracks ,uncoated areas and other detects .
No visible laps, seams, cracks or bursts.

4.厚度检测 Thickness Measurement

用测厚仪在同炉试样上进行检测,经检测厚度为0.0031-0.0033in。

The thickness was measured on test strip by thickness testing machine. It was 0.0031-0.0033in

5.成分分析 Composition Analysis

成份检测在同炉试样中进行,经检测磷的含量约为10.54%,其余为镍含量。

The chemical composition for the deposit is 10.54% phosphorous and balance is nickel.

6.硬度检测 Hardness Measurement

经显微硬度计检测,其镀层硬度为:750-800HV₁₀₀

The hardness of the ENP coating was tested by a microhardness tester and found to be 750-800HV₁₀₀

7.试片在室温下浓度为70%的硝酸溶液中浸泡30秒。未发现试片表面有变黑等现象。

The test strip immersed in 70% nitric acid solution at ambient temperature for 30 seconds ,and didn't turn black .

结论Conclusion 合格(√)Conformity 不合格()Non-Conformity

检验员
Inspector

caojuhe

质保经理
Q.A Manager



Stem: 122432354002

供应商名称 Supplier Name		检验证明书-质量控制部 Inspection Certificate-Quality Control Dept.						ISO9001公司质量管理体系证书号 Company's QMS Certificate No.															
南京肯特复合材料有限公司 Nanjing CompTECH Composites Corporation		证书号 / Cert. No.		日期 / Date		CN17/21222.00																	
		10008FG251953-49		2025/10/10																			
买方 Purchaser		订单号 Purchase Order No.				证书类型 Certificate Type: EN 10204																	
		4514912202				V 3.1 3.2 2.2																	
山东卡麦龙石油机械有限公司 Shandong Cameron Petroleum Equipment Pte Ltd		工单号 Our Job No.				熔炼方法 Melting Practice																	
		10008FG251953-49				V EF																	
化学分析 Chemical Analysis						测试标准 Test Specification: ASTM A751																	
材料规范 MS Code Material Specification		炉号 Heat No.		3301711		原材料供应商 Raw Material Supplier																	
		热处理批次号 H.T.Lot No.		240324H04		Shandong Shouguang Juneng																	
MS-021272-01(03)		C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	B	V	Nb	N	Nb+Ta						
		%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%						
最小 Min.		0.38	0.60			0.90					0.15												
最大 Max.		0.45	0.90	0.40	0.50	1.20	0.035	0.035		0.35	0.25			0.05									
浇包分析 Heat Analysis		0.42	0.89	0.25	0.03	0.99	0.007	0.011		0.04	0.200			0.007									
产品分析 Product		0.43	0.880	0.26	0.01	0.97	0.009	0.013		0.02	0.21			0.007									
碳当量 CE (Long)=				Max.:		Cr+Mo=		Max.:		Cu+Ni+Cr+Mo+V=						Max.:							
热处理 Heat Treatment																							
类型 Type		温度 Temperature (°C)				保温时间 Keep Time (Hour)				冷却方法 Cooling Method													
淬火 (Quenching)		880				2				聚合物(Polymer)													
回火 (Tempering)		680				3				空冷(Air Cooling)													
机械性能测试 Mechanical Test				测试标准 Test Specification ASTM A370						Test Data From QTC / 试棒													
材料规范 MS Code		抗拉强度 Tensile Strength Mpa		屈服强度 Yield Strength 0.2% offset Mpa		延长率 Elongation (%)		收缩率 Reduction of Area (%)		硬度 Hardness HBW/P 10/3000		冲击测试 Impact Test											
MS-021272-01(03)												方向 Diret.	类型 Type	温度 °C Temp	平均 J Avg.	最小 J Min.	Actual J 实测						
最小 Min.		677		520		15		45		HBW		L	V	-46	27	27	128						
最大 Max.										237							108						
实测 Actual		705		546		33		71		224 228 227							118						
试样号 Test Piece No.		标距 mm Gauge Length		直径 mm Diameter		缩颈 mm Length Of Reduced Section		温度 °C Temp		方向 Direction													
3301711 240324H04		50		12.5		60		室温 Room Temp		L													
本体硬度 Hardness From Piece HBW/P 10/3000		210		228		Remark: At Least Two Pieces Per Batch When The Batch Is Greater Than One Piece. 当批次大于1件时, 每批次至少检2件.																	
镀层分析 Analysis For Nickel Plating																							
电镀规范 (需要填) Plating Specification		要求厚度 Inch Required Thickness		实测厚度 Inch Actual Thickness		热处理温度 °C Heat Treatment Temperature		热处理时间 小时 Heat Treatment Time Hour															
SP-054677-01-01-75(03)		0.003		0.003																			
订单行号 PO Item No.		锻造比 Forging Ratio		描述 Description				部件号 Part Number				数量 Quantity											
570		4:1		STEM B5 75 MM 42CrMo4 3 MIL ENP				2604929-75-07-02(02)				3											
备注 Remarks:																							
材料符合规范 Material According To The Spec		NACE MR0175/ISO15156				42CrMo4 UNI EN 10083-1																	
热处理校验符合API 6D 25th. Heat Treatment Furnaces Calibrated According To API 6D 25th Ed.																							
在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明, 材料没有被放射性污染。 We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise stated, material free from radioactive contamination.																							
检查员签字/日期 Surveyor's Name, Signature & Date						审核员签字日期 Inspector's Name, Signature & Date																	
Zaiqin Chen						2025/10/10						Weiming Sun						2025/10/10					





聚恩辈

温州聚恩辈实业有限公司检测报告



JUENBEI INDUSTRIAL WENZHOU JUENBEI INDUSTRIAL CO.,LTD. ENP INSPECTION REPORT

No. MS252744-01 R0

客户名称 Customer Name	南京肯特复合材料 股份有限公司	产品类型 Production Name	STEM	报告编号 Report No.	02510016471-06
订单号\行号 P.O. No.\Item No.	4514912202/00570	材料+电镀厚度 Material+ENP Thickness	42CrMo4 UNI EN 10083-1+0.075	日期 Inspection Date	2025-10-13
型号规格 Production Size	2604929-75-07-02	数量 Production Quantity	3PCS	热处理温度/时间 Heat/Time	N/A
硬度要求 Hardness Require	≥500HV	电镀标准 Carry out Standard	SP-054677-01-01-75 03	电镀批号 Batch No.	02510016471
检测记录 Inspection Record					
1. 附着力测试 Adhesion Test					
附着力测试根据ASTM B571标准在试样上进行,将试样弯曲180°后,未发现镀层有剥落现象。 Adhesion test as per ASTM B571 the sample was bend at 180 degree over a Mandrel of Same size as sample No peeling was observed.					
2. 孔隙测试 Porosity Test					
根据ASTM B733 9.6.1标准进行孔隙试验,未发现蓝色斑点。 Ferroxyl test carried out as per ASTM B733 standard to check for porosity None were found.					
3. 外观检测Visual Test					
外观检测涂层光滑、粘附,无可见的水泡、凹坑、结节、气孔、裂缝、未涂层区域和其他缺陷。 The appearance inspection shows that the coating is smooth and adherent, without visible blisters, pits, nodules, pores, cracks, uncoated areas and other defects.					
4. 厚度检测Thickness Test					
用测厚仪在同炉实物上进行检测,经检测厚度为0.076-0.079mm The thickness was measured on matter and was found to be 0.076-0.079mm With Thickness Testing Machine.					
5. 成份检测Composition Test					
成份检测在同炉试样中进行,经检测磷的含量约为10.29%,其余为镍含量。 The job were plated with E.N.P having average phosphorus content of 10.29% and Remaining being nickel.					
6. 硬度检测Hardness Test					
产品电镀后检测其平均硬度值为527HV100 Hardness after plating average527HVat 100 gm load.					
7. 硝酸试验NITRIC ACID TEST					
样板浸入硝酸中30秒,未发现发黑。 The sample plate is immersed in nitric acid for 30 seconds, No blackening found .					
结论Conclusion	合格(√)Conformity 不合格()Non-Conformity				
检验员 Inspector:	20251013		审 批 Approved by:	袁文亮	



Seats: 122432354002

YL IX/II-17-6.7 REV.B/0

JIANGSU YINGLIU MACHINE MANUFACTURING CO., LTD. 江苏应流机械制造有限公司				INSPECTION CERTIFICATE QUALITY CONTROL DEPARTMENT 检验证明书 (质量控制部)				Company's QMS certified body—ISO9001 No公司质量管理体系证书号: No29333-2008-AQ-RGC-RvA/Rev.1												
Purchaser: 买方 Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦石油机械有限公司				Cert. No. 证书号:	25101908		Date 日期:	2025. 10. 28		Material Specification 材料规格:		MS-021239-02-04 REV.03								
				Purchase Order No. 订单号:		4514889827				Certificate type 证书类型:		EN 10204 □2.2. ■3.1 □3.2								
				Our Job No. 工单号:		YC27SH210				Melting practice 熔炼方法:		■E.F. □E.F.V.D.								
CHEMICAL ANALYSIS 化学分析												Test Specification 测试标准: ASTM A751								
Material 材料规格:		ASTM A350 GRADE LF2 CLASS 1		Heat No. 炉号:		25-110307		H.T.Lot 热处理批次号:		251017-A1G1			Raw Material Supplier 原材料供应商:			CHANG QIANG				
MS-021239-02-04 REV.03		C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	H			
		%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm			
Min			0.57	0.13																
Max		0.23	1.39	0.32	0.43	0.33	0.02	0.025		0.43	0.13			0.03	0.02					
Heat Analysis 热分析		0.200	0.930	0.230	0.0300	0.070	0.0060	0.0180		0.010	0.0035			0.0005	0.0010					
Product Analysis 产品分析		0.205	0.965	0.232	0.0105	0.0722	0.0036	0.0147		0.0024	0.0035			0.00050	0.0010					
碳当量C.E. (Long) (≤0.43)		0.382				Cr+Mo (≤0.32)		0.076		Cu+Ni+Cr+Mo+V (≤1.00)			0.089			PREN		-		
HEAT TREATMENT 热处理 Heat treatment equipment comply with API 6D 25 edition and API 6A 21 edition.																				
Type 类型		Temperature (°C) 温度								Time (Hour) 时间				Cooling Method 冷却方法						
Quenching		895								1.73h				Water cooled						
Tempering		610								2.50h				Air cooled						
MECHANICAL TEST 机械性能测试		Test Specification 测试标准: ASTM A370-24a								From QTC 试棒: ■ (100mm*100mm*200mm)				From Piece 本体: □						
Location Direction 位置/方向		UTS σ b 抗拉强度		Ys σ s (0.2% offset) 屈服强度		EL δ 5 伸长率		ROA φ 收缩率		IMPACT TEST 冲击测试 10x10x55 mm						HARDNESS (HBW) 硬度				
1/2 T longitudinal		(Mpa)		(Mpa)		(%)		(%)		Location Direction 位置/方向		Type 类型		Temp °C 温度		Min 最小		Avr 平均	Min 最小	Max 最大
Min		485		277		22		30		1/2 T longitudinal		V		-46°C		27		27	143	197
Max		655		/		/		/												
ACTUAL 实测		UTS σ b 抗拉强度		Ys σ s (0.2% offset) 屈服强度		EL δ 5 伸长率		ROA φ 收缩率		IMPACT TEST 冲击测试 10x10x55 mm						HARDNESS (HBW) 硬度				
		(Mpa)		(Mpa)		(%)		(%)		Location Direction 位置/方向		Type 类型		Temp °C 温度				Joules 焦耳		
		564		342		35		73		1/2 T longitudinal		V		-46°C				136 147 155 146		160/158/159
Yield/Tensile strength ratio 屈服抗拉强度比				-				Lateral expansion (mm) 侧膨胀				-								
Test Piece No. 试样号		Gauge length 标距 (mm)		Diameter 直径 (mm)		length of reduced section 缩颈 (mm)				Temp 温度 (°C)				Direction 方向						
10-219#		50.00		12.56		60				20				longitudinal 纵向						
From Piece 本体 HARDNESS 硬度 (HBW/P 10/3000)		160		158		Remark: At least two pieces per batch when the batch is greater than one piece 当批次大于1件时, 每批次至少检两件														
Analysis for Plating 镀层分析																				
Plating specification 电镀规范				Thickness 厚度 (in)		Heat Treatment Temperature 热处理温度 (°C)				Heat Treatment time 热处理时间 (h)				Surface Hardness 产品表面硬度 (HV100)						
SP-054677-01-01-75 REV.03		Standard 标准		0.003+0.0005 /-0		/				/				Min 最小 500 Max 最大 /						
		Actual 实测		0.003		/				/				500						
PO Item No. 订单行号		Your Item No. 你们的行号				Description 描述				Forging ratio 锻造比		Part Number 部件号		Quantity 数量						
40						OUTER SEAT RING B5 8" 150 - 900				4.6:1		2151914-02-02		40						
110						SEAT ASSY B5R 8" 900 LF2 3MIL				4.5:1		2645459-08-09-03		4						
Remarks 备注:																				
Material according to the spec: 材料符合规范:				NAME: NR0175/ISO15456				ASTM A350 GRADE LF2 CLASS 1												
Heat treatment furnaces calibrated according to API 6D 25th Ed. Annex F and API 6A 21th Ed. Annex M 热处理炉符合 API 6D 25th 版附录 F 和 API 6A 21th 附录 M。 We hereby certified that the items described herein have been manufactured and inspected in accordance with the specifications and requirements of bill of material called for against the above order unless otherwise is stated. Material free from radioactive contamination 在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明, 材料没有被放射性污染。																				
Prepared by 编制签字:		李萍				Proofread by 校对员签字:				张静				Inspector's Name, Signature 审核员签字:		张静				

APPROVED

GAndamuthu, 10-01-2026, 12:15:19



青田永拓金属表面处理有限公司

Qingtian Yongtuo Metal Surface Technical Treatment Co., Ltd.

化学镀镍磷检测报告 ENP INSPECTION REPORT

日期 Date: 2025-11-09

客户名称 Customer Name	江苏应流机械制造有限公司 限责任公司	产品类型 Production Name	阀座组件 SEAT ASSY	报告编号 Report No.	251109-2
订单号\行号 P.O. NO. \Item NO.	4514889827/110	材料+电镀厚度 Material+ENP Thickness	A350 LF2 CL1 +ENP0.075+ERTALON 6XAU	跟踪卡号 Route NO	/
型号规格 Production Size	08-0900 9G-P22D08909	数量 Production Quantity	4件	电镀批号 Batch No	YT15013
硬度要求 Hardness Require	≥500	电镀标准 Carry out Standard	SP-054677-01-01- 75 (Rev06/03)	后热处理温度/时间 Post Heat Treatment.	N/A
检测记录 Inspection Record					
1. 附着力测试 Adhesion Test 附着力测试根据ASTM B571标准在试样上进行, 将试样弯曲180° 后, 未发现镀层有剥落现象。 Adhesion test as per ASTM B571 the sample was bend at 180 degree over a Mandrel of Same size as sample No peeling was observed.					
2. 孔隙测试 Porosity Test 根据ASTM B733 9.6.1标准进行孔隙试验, 未发现蓝色斑点。 Ferroxyl test carried out as per ASTM B733 standard to check for porosity None were found.					
3. 外观检测 Visual Test 外观检测发现表面光滑并粘附, 未发现表面存在水泡、凹坑、结节、气孔、裂纹等缺陷。 Appearance inspection found that the surface was smooth and adhered, and no surface defects such as blisters, pits, nodules, pores and cracks were found.					
4. 厚度检测 Thickness Test 用测厚仪在同炉实物上进行检测, 经检测厚度为0.075-0.080mm The thickness was measured on matter and was found to be 0.075-0.080mm With Thickness Testing Machine.					
5. 成份检测 Composition Test 成份检测在同炉试样中进行, 经检测磷含量约为10.80%, 镍的含量约为89.20%。 The content of phosphorus and nickel is about 10.80% and 89.20% respectively.					
6. 硬度检测 Hardness Test 产品电镀后检测其平均硬度值为:550HV100。 Hardness after plating average :550HVat 100 gm load					
7. 硝酸实验检测 Nitric acid test test 将镀有75微米化学镀镍的试样条侵入70%的硝酸溶液中静放30秒, 取出后, 表面未发现色差。 The sample plated with 75 micron chemical nickel plating was inserted into the 70% nitric acid solution for 30 seconds. After taking out, no color difference was found on the surface.					
结论 Conclusion	合格(√)Conformity 不合格()Non-Conformity			 单位盖章 Stamp of organization	
检验员: Inspector	项余梅	审批: Approved by	练徐波		

VI IX/II-17-6.7 REV.B/0

JIANGSU YINGLIU MACHINE
MANUFACTURING CO., LTD.
江苏应流机械制造有限公司INSPECTION CERTIFICATE
QUALITY CONTROL DEPARTMENT
检验证明书 (质量控制部)Company's QMS certified body—ISO9001
No公司质量管理体系证书号: No29333-2008-AO-RGC-RvA/Rev.1

Purchaser: 买方 Shandong Cameron Petroleum Equipment Pte Ltd 山东卡麦龙石油机械有限公司	Cert. No. 证书号:	25090506	Date 日期:	2025. 09. 08	Material Specificaiton 材料规范:	MS-021239-02-04 REV. 03
	Purchase Order No. 订单号:	4514851272			Certificate type 证书类型:	EN 10204 ☐ 2.2. ☒ 3.1 ☐ 3.2
	Our Job No. 工单号:	YG27SD03B			Melting practice 熔炼方法:	☒ E.F. ☐ E.F.V.D.

CHEMICAL ANALYSIS 化学分析									Test Specification测试标准：ASTM A751								
Material 材料规范:	ASTM A350 GRADE LF2 CLASS 1		Heat No. 炉号:		5204125		H.T.Lot 热处理批次号:		250903-A101			Raw Material Supplier 原材料供应商:			DONG FANG TE GANG		
MS-021239-02-04 REV.03	C	Mn	Si	Ni	Cr	S	P	Al	Cu	Mo	Ti	Sn	V	Nb	N	H	
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	
Min		0.57	0.13														
Max	0.23	1.39	0.32	0.43	0.33	0.02	0.025		0.43	0.13			0.03	0.02			
Heat Analysis 浇包分析	0.200	0.960	0.240	0.0100	0.120	0.006	0.0160		0.020	0.0010			0.0020	0.0010			
Product Analysis 产品分析	0.197	0.978	0.226	0.0067	0.1280	0.0044	0.0174		0.0033	0.0014			0.00180	0.0057			
碳当量C.E. (Long) (≤0.43)			0.387		Cr+Mo (≤0.32)		0.129		Cu+Ni+Cr+Mo+V (≤1.00)			0.141		PREN		-	

HEAT TREATMENT 热处理			
Heat treatment equipment comply with API 6D 25 edition and API 6A 21 edition.			
Type 类型	Temperature (°C) 温度	Time (Hour) 时间	Cooling Method 冷却方法
Quenching	895	1.50h	Water cooled
Tempering	620	2.00h	Air cooled

MECHANICAL TEST 机械性能测试		Test Specification 测试标准: ASTM A370-24a			From QTC 试棒: ■ (100mm*100mm*200mm)			From Piece 本体: □					
Location Direction 位置/方向	UTS σ b 抗拉强度	Ys05 (0.2% offset) 屈服强度	EL 65 伸长率	ROA ϕ 收缩率	IMPACT TEST 冲击测试 10x10x55 mm							HARDNESS (HBW)硬度	
1/2 T longitudinal	(Mpa)	(Mpa)	(%)	(%)	Location Direction 位置/方向	Type 类型	Temp °C 温度	Min 最小		Avr 平均	Min 最小	Max 最大	
Min	485	277	22	30	1/2 T longitudinal	V	-46°C	27		27	143	197	
Max	655	/	/	/									
ACTUAL 实测	UTS σ b 抗拉强度	Ys05 (0.2% offset) 屈服强度	EL 65 伸长率	ROA ϕ 收缩率	IMPACT TEST 冲击测试 10x10x55 mm							HARDNESS (HBW)硬度	
	(Mpa)	(Mpa)	(%)	(%)	Location Direction 位置/方向	Type 类型	Temp °C 温度	Joules 焦耳					
	547	357	32	76	1/2 T longitudinal	V	-46°C	154	122	156	144	163/160/159	
Yield/Tensile strength ratio 屈服抗拉强度比			-		Lateral expansion(mm) 侧膨胀			-					
Test Piece No. 试样号		Gauge length 标距 (mm)	Diameter 直径 (mm)	length of reduced section 缩颈 (mm)			Temp 温度 (°C)			Direction 方向			
9-26#		50.00	12.52	60			20			longitudinal 纵向			
From Piece本体 HARDNESS 硬度 (HBW/P 10/3000)		163	160	Remark: At least two pieces per batch when the batch is greater than one piece 当批次大于1件时, 每批次至少检两件									

Analysis for Plating 镀层分析					
Plating specification 电镀规范		Thickness 厚度 (in)	Heat Treatment Temperature 热处理温度 (°C)	Heat Treatment time 热处理时间 (h)	Surface Hardness 产品表面硬度 (HV100)
/	Standard 标准	/	/	/	Min 最小
	Actual 实测	/	/	/	Max 最大

PO Item No. 订单行号	Your Item No. 你们的行号	Description 描述	Forging ratio 锻造比	Part Number 部件号	Quantity 数量
30		BODY B4D 02 150 A350 LF2	4.1:1	2605464-02-01-01	10
40		CLOSURE B4D 03 300 RF A350 LF2	4.2:1	2605468-03-03-01	8
50		CLOSURE B4D 02 150 RF A350 LF2	4.4:1	2605468-02-01-01	18
130		BODY B5 6" 600	4.4:1	50374544	7
396		LOWER COVER PLATE 2 15G/5M 3 6C/9C 4 150	4.1:1	2365374-20-26	9
500		TOP COVER 75 MM A350 LF2	4.4:1	2604931-75-01-01	24

Remarks 备注:			
Material according to the spec:		API 6D 25th Ed. Annex F and API 6A 21th Ed. Annex M 符合 API 6D 25th 版附录 F 和 API 6A 21th 附录 M.	ASTM A350 GRADE LF2 CLASS 1
材料符合规范:			
Heat treatment furnaces calibrated according to API 6D 25th Ed. Annex F and API 6A 21th Ed. Annex M 热处理炉符合 API 6D 25th 版附录 F 和 API 6A 21th 附录 M.			
We hereby certified that the items described herein have been manufactured and inspected to conform with the specifications and requirements of bill of material called for against the above order unless otherwise stated. Material free from radioactive contamination 在此声明我们以上订单所描述的产品均按照订单中材料清单所含规范和要求进行制造和检验, 除非另有说明, 材料没有放射性污染。			
Prepared by 编制签字:	李库	Inspector's Name, Signature 审核员签字:	尹晓莹



Shandong Cameron Petroleum Equipment Pte Ltd
No 2577 Gaoxin Sh Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0801

TEST CERTIFICATE FOR GROVE B5 PRODUCT IN ACCORDANCE WITH EN 10204 3.1

Customer Name: SCHLUMBERGER CANADA LIMITED
CC STO No./PO No.: 4710718222
Sales Order reference number: N/A
Item number: 30
Work order number: 122352167
Valve Part Number: 2510873-08-08-03
Valve Serial Number: 122352167001
Tag Number: N/A

All valves have been inspected and hydrostatically tested to comply with the specifications and corresponding test parameters indicated below and are certified as satisfactory.

This certification complies with EN 10204 para. 3.1. It was prepared by QA department personnel from tests conducted by a testing department whose supervisor is administratively independent from the Production Manager.

Hydrostatic shell & seat tests conducted in accordance with API 6D

Valve size: 8"X8"
Material specification: ASTM A350 LF2

ANSI pressure class: 900
CWP rating: 2220 psig / 15.32 Mpa

Hydrostatic Shell & Seat Test

Shell test pressure:	3330	psig	/	22.98	Mpa	Shell test duration:	5	minutes
Seat 'A' test pressure:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
Seat 'B' test pressure:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
Double Block & Bleed (DBB) test press.:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
Double Isolation & Bleed (DIB-1) test press.:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
The connection test for Relief Valve:	2871	psig	/	18.40	Mpa	Shell test duration:	5	minutes

Pressure gage No. 压力表编号	PN207	PN256	PN271	
Due date 有效日期	2025.8.10	-	-	

Note: Pressure Listed above is minimum required. Maximum pressure is not allowed to exceed 1.1~1.3 times listed above

Acceptance Criteria: No visually detectable leakage for the duration of the test (ISO5208 Rate A)

Actuator Type: -

Actuator Brand: -

Actuator Figure No.: -

Configuration DPEXDPE

Additional information for valve

Pressure Test Inspector:

Name: Zhang Jinyang

Signature: Zhang Jinyang

Date: 2025.6.18

QA/QC Approval:

Name: Li Liang

Signature: Li Liang

Date: 2025.6.18

Third Party Witness

Name:

Signature:

Date:



Shandong Cameron Petroleum Equipment Pte Ltd
No 2577 Gaoxin 5th Road, Gaoxin District, Weihang City
Shandong Province, P.R. China, Postcode 251041
Tel/Fax: +86-536-203-0801

TEST CERTIFICATE FOR GROVE B5 PRODUCT IN ACCORDANCE WITH EN 10204 3.1

Customer Name: SCHLUMBERGER CANADA LIMITED
CC STO No./PO No. 4710718222
Sales Order reference number: N/A
Item number: 30
Work order number: 122352167
Valve Part Number: 2510873-08-09-03
Valve Serial Number: 122352167001
Tag Number: N/A

All valves have been inspected and low pressure gas tested to comply with the specifications and corresponding test parameters indicated below and are certified as satisfactory.

This certification complies with EN 10204 para. 3.1. It was prepared by QA department personnel from tests conducted by a testing department whose supervisor is administratively independent from the Production Manager.

Low gas seat tests conducted in accordance with API 6D

Valve size: 8"X8"	ANSI pressure class: 900
Material specification: ASTM A350 LF2	CWP rating: 2220 psig / 15.32 Mpa
<u>Air Seat Test</u>	
Air seat 'A' test pressure: 80 psig / 0.55 Mpa	Seat test duration: 5 minutes
Air seat 'B' test pressure: 80 psig / 0.55 Mpa	Seat test duration: 5 minutes
Double Block & Bleed (DBB) test press.: 80 psig / 0.55 Mpa	Seat test duration: 5 minutes

Pressure gage
No. 压力表编号

Due date
有效日期

Note:

Pressure Listed above is minimum required, Maximum pressure is not allowed to exceed 100psi.

Acceptance Criteria: No visually detectable leakage for the duration of the test (ISO5208 Rate A)

Actuator Type: -

Actuator Brand: -

Actuator Figure No.: -

Configuration DPEXDPE

Additional information for valve

Pressure Test Inspector:

Name: Zhang Qingyue

Signature: [Signature]

Date: 2015.6.18

QA/QC Approval:

Name: Li Liang

Signature: [Signature]

Date: 2015.6.18

Third Party Witness

Name: _____

Signature: _____

Date: _____



Shandong Cameron Petroleum Equipment Pte Ltd
No 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax +86-536-208-0601

TEST CERTIFICATE FOR GROVE B5 PRODUCT IN ACCORDANCE WITH EN 10204 3.1

Customer Name: STAFFORD - 13127 TRINITY DR
CC STO No./PO No.: 4710753518
Sales Order reference number: N/A
Item number: 40
Work order number: 122432354
Valve Part Number: 2510873-08-09-03
Valve Serial Number: 122432354002
Tag Number: N/A

All valves have been inspected and hydrostatically tested to comply with the specifications and corresponding test parameters indicated below and are certified as satisfactory.

This certification complies with EN 10204 para. 3.1. It was prepared by QA department personnel from tests conducted by a testing department whose supervisor is administratively independent from the Production Manager.

Hydrostatic shell & seat tests conducted in accordance with API 6D

Valve size: 8"X8" ANSI pressure class: 900
Material specification: ASTM A350 LF2 CWP rating: 2220 psig / 15.32 Mpa

Hydrostatic Shell & Seat Test

Shell test pressure:	3330	psig	/	22.98	Mpa	Shell test duration:	5	minutes
Seat 'A' test pressure:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
Seat 'B' test pressure:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
Double Block & Bleed (DBB) test press.:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
Double Isolation & Bleed (DIB-1) test press.:	2442	psig	/	16.85	Mpa	Seat test duration:	5	minutes
The connection test for Relief Valve:	2671	psig	/	18.40	Mpa	Shell test duration:	5	minutes

Pressure gage
No. 压力表编号

PM364

PM365

PM368

Due date
有效日期

2015-12-6

Note: Pressure Listed above is minimum required. Maximum pressure is not allowed to exceed 1.1~1.3 times listed above

Acceptance Criteria: No visually detectable leakage for the duration of the test (ISO5208 Rate A)

Actuator Type: -

Actuator Brand: -

Actuator Figure No.: -

Configuration DPEXDPE

Additional information for valve

Pressure Test Inspector:

Name: gang chun yuam

Signature: [Signature]

Date: 2015-12-6

QA/QC Approval:

Name: Li Liang

Signature: [Signature]

Date: 2015.12.6

Third Party Witness

Name: _____

Signature: _____

Date: _____



Shandong Cameron Petroleum Equipment Pte Ltd
No 2577 Gaoxin 5th Road, Gaoxin District, Weifang City
Shandong Province, P.R. China, Postcode 261041
Tel/Fax: +86-536-208-0601

TEST CERTIFICATE FOR GROVE B5 PRODUCT IN ACCORDANCE WITH EN 10204 3.1

Customer Name: STAFFORD - 13127 TRINITY DR
CC STO No./PO No. 4710753518
Sales Order reference number: N/A
Item number: 40
Work order number: 122432354
Valve Part Number: 2510873-08-09-03
Valve Serial Number: 122432354002
Tag Number: N/A

All valves have been inspected and low pressure gas tested to comply with the specifications and corresponding test parameters indicated below and are certified as satisfactory.
This certification complies with EN 10204 para. 3.1. It was prepared by QA department personnel from tests conducted by a testing department whose supervisor is administratively independent from the Production Manager.

Low gas seat tests conducted in accordance with API 6D

Valve size: 8"X8"

ANSI pressure class: 900

Material specification: ASTM A350 LF2

CWP rating: 2220 psig / 15.32 Mpa

Air Seat Test

Air seat 'A' test pressure: 80 psig / 0.55 Mpa Seat test duration: 5 minutes

Air seat 'B' test pressure: 80 psig / 0.55 Mpa Seat test duration: 5 minutes

Double Block & Bleed (DBB) test press.: 80 psig / 0.55 Mpa Seat test duration: 5 minutes

Pressure gage

No. 压力表编号

PM351

Due date

有效日期

2025-12-6

Note: Pressure Listed above is minimum required, Maximum pressure is not allowed to exceed 100psi.

Acceptance Criteria: No visually detectable leakage for the duration of the test (ISO5208 Rate A)

Actuator Type: -

Actuator Brand: -

Actuator Figure No.: -

Configuration DPEXDPE

Additional information for valve

Pressure Test Inspector:

Name: gong chun gao

Signature: [Signature]

Date: 2025-12-6

QA/QC Approval:

Name: Li Liang

Signature: [Signature]

Date: 2025.12.6

Third Party Witness

Name: _____

Signature: _____

Date: _____



BALL VALVE TEST RECORD

Date: 7/1/26WO#: Q35762-1

F-583 017, R6

CUST: Energy EquipmentPO#: 260128-2SPEC: ConsumersSIZE/CLASS/TYPE: 8" 900 GroveEND CONNECTION: RFFEVALVE SER #: 122352167001

TEST MEDIUM: ETHYLENE GLYCOL / WATER SOLUTION

ACTUATOR TYPE: —ACT. SER. #: —CHART ID #: 3674GAGE ID #: 59

HYDROSTATIC TEST

SHELL TEST: 3450 psi DURATION: 15 minTORQUE @ FULL DIFF/MAOP: LEFT: — lb-ft RIGHT: — lb-ft MAX: — lb-ftDBB: — lb-ft DIB1: — lb-ftBREAKOUT @ Full DIFF/MAOP:
(Circle one)LEFT: PASS / FAIL NA RIGHT: PASS / FAIL NADBB: PASS / FAIL NA DIB1: PASS / FAIL NALEFT SEAT: 2445 psiDURATION: 5 minRIGHT SEAT: 2445 psiDURATION: 5 minDBB: 2445 / 2445 psiDURATION: 5 / 5 min ^{open} _{closed}DIB-1: 2445 psiDURATION: 5 minAll Lube Lines Tested @ — psi for — min.

GAS / AIR TEST

AIR SEAT LEFT: — psiDURATION: — minAIR SEAT RIGHT: — psiDURATION: — minDBB AIR: — psiDURATION: — minDIB-1: — psiDURATION: — min

By signature QTR certifies the above valve has satisfactorily passed all test indicated on this record.

-No visible leakage or inelastic deformation were evident during shell test

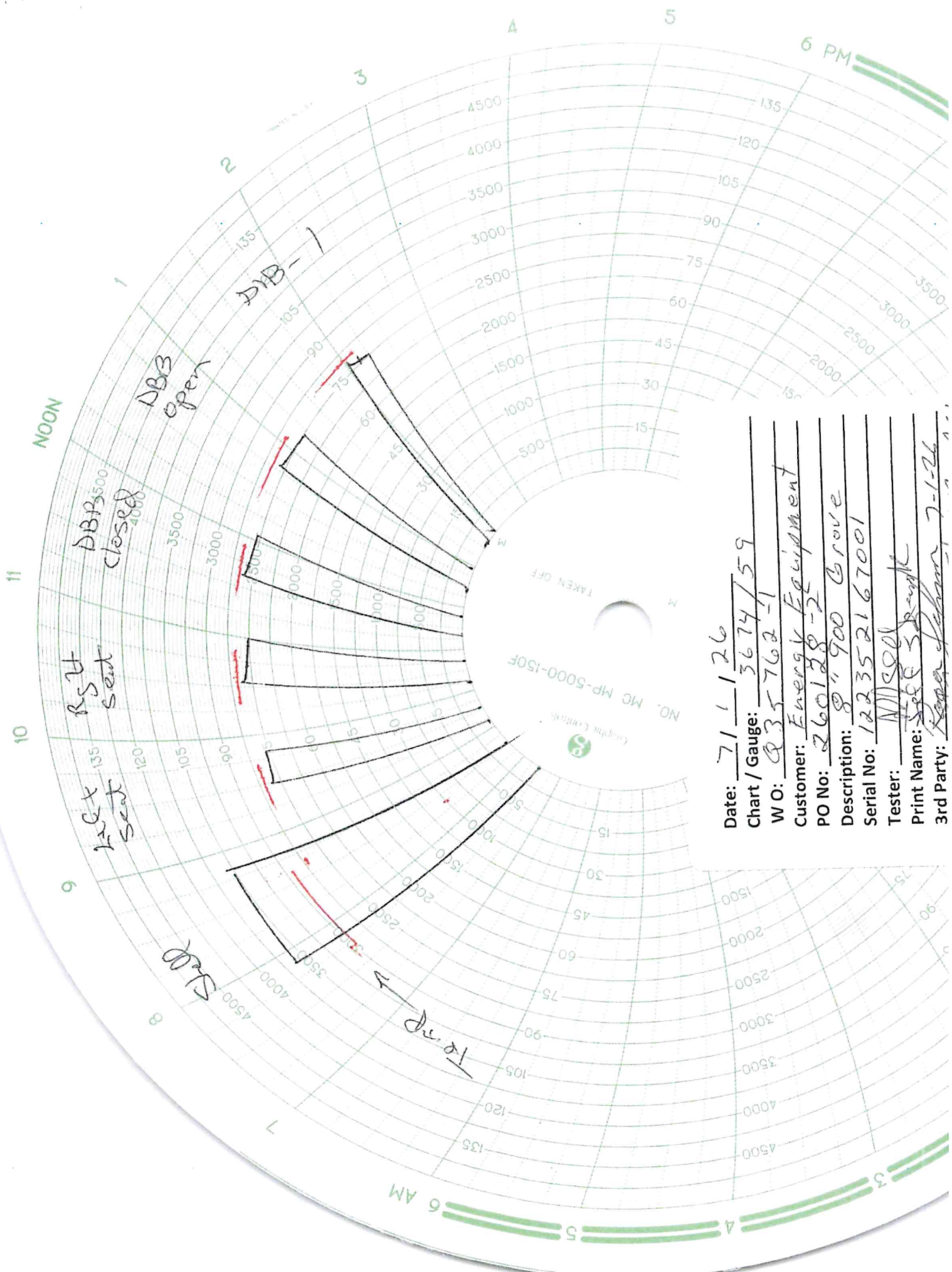
-No visible leakage evident during seat testing per ISO 5208 rate A

QTR

Date: 7/1/26Print: Jeff Schrock

INSP (if Req'd):

Date: 7/1/26Print: Roger Johnson AC1



Date: 7/1/26
 Chart / Gauge: 3674/59
 W O: Q35762-1
 Customer: Energy Equipment
 PO No: 260138-2
 Description: 8" 900 Grove
 Serial No: 122352167001
 Tester: MISOL
 Print Name: Mike Smith
 3rd Party: Robert Johnson 7-1-26



BALL VALVE TEST RECORD

Date: 7/1/26
WO#: Q35762-1

F-580-017, R6

CUST: Energy Equipment PO#: 260128-2
SPEC: Consumers
SIZE/CLASS/TYPE: 8" 900 Grove END CONNECTION: RFFE
VALVE SER #: 122432354002 TEST MEDIUM: ETHYLENE GLYCOL / WATER SOLUTION
ACTUATOR TYPE: — ACT. SER. #: —
CHART ID #: 11694158005 GAGE ID #: 56

HYDROSTATIC TEST

SHELL TEST: 3450 psi DURATION: 15 min.
TORQUE @ FULL DIFF/MAOP: LEFT: — lb-ft RIGHT: — lb-ft MAX: — lb-ft
DBB: — lb-ft DIB1: — lb-ft
BREAKOUT @ Full DIFF/MAOP: LEFT: PASS / FAIL NA RIGHT: PASS / FAIL NA
(Circle one) DBB: PASS / FAIL NA DIB1: PASS / FAIL NA
LEFT SEAT: 2445 psi DURATION: 5 min
RIGHT SEAT: 2445 psi DURATION: 5 min
DBB: 2445 / 2445 psi DURATION: 5 / 5 min open
DIB-1: 2445 psi DURATION: 5 min closed

All Lube Lines Tested @ — psi for — min.

GAS / AIR TEST

AIR SEAT LEFT: — psi DURATION: — min.
AIR SEAT RIGHT: — psi DURATION: — min.
DBB AIR: — psi DURATION: — min.
DIB-1: — psi DURATION: — min.

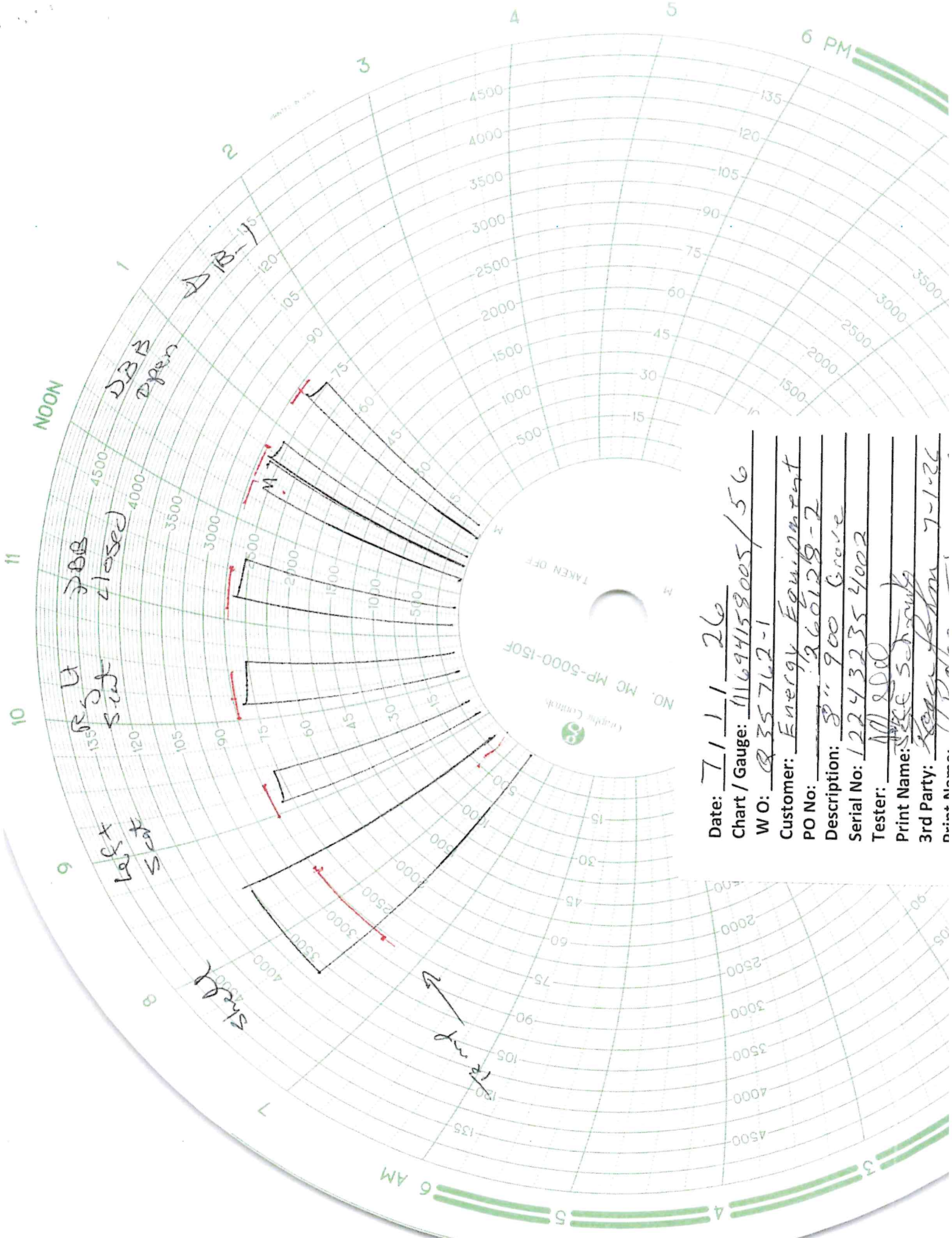
By signature QTR certifies the above valve has satisfactorily passed all test indicated on this record.

-No visible leakage or inelastic deformation were evident during shell test

-No visible leakage evident during seat testing per ISO 5208 rate A

QTR [Signature] Date: 7/1/26
Print: Jeff Schreck

INSP (if Req'd): [Signature] Date: 7/1/26
Print: Robert Johnson ACI



Date: 7/1/1 26
Chart / Gauge: 11694158005/56
WO: Q35762-1
Customer: Energy Equipment
PO No: 260128-2
Description: 8" 900 Grove
Serial No: 122432354002
Tester: W.D.D.
Print Name: W.D.D.
3rd Party: 7-1-26

Customer: Energy Equipment PO# 260128-2
 Q# 35762-1 WI# 136-F-003 (Consumers Green)
 QTY# 2 Description: 8" 900# RFFE Grove B5 Ball Valve, G/O w/HW

Surface Preparation

Surface free of Contamination:	Yes	Solvent Cleaned:	SSPC-SP-1
Soluble Salt test:	Yes	Salt test performed:	Potassium Ferricyanide
Blast Pressure:	110 psi	Test Plate:	On Request
Blotter Test:	ASTM D 4285	Abrasive Type:	Medium
Surface Profile Standard:	SSPC-SP-1	Blast Acceptance:	SP-10
Profile required	2.4mils	Profile Average:	2.85
Testex Tape Sheet Attached:	NA	Blast time:	7-15-26 8:15am

Blast profile Tape



Coating Conditions

Coat	RH%	TA°	TS°	TD°	TsTd°	Date	Time
1st	50.1	90.2	88	69.1	18.9	7/15/2026	12:40pm
2nd	65	82.1	83.1	69.1	14	7/16/2026	8:00am
3rd							

Coating Materials

COAT	MANUFACTURER	PAINT	COLOR	BATCH A#	BATCH B#	Temp
1st	Sherwin Williams	Duar-Plate	White	XM3175RL	XM3305CA	78°
2nd	Sherwin Williams	Hi-Solids Poly	Consumers Grn	XM1176BJ	MQ1126TG	77°
3rd						

Coating DFT

SN#	Stripe	1st Coat	2nd Coat	3rd Coat	4th Coat	Total
122352167001		9.77	13.3			13.3
122432354002		9.5	14.62			14.62

Q35762-1

FINAL INSPECTION

Customer Specification #

Consumers

RESULTS

DFT Avg.	Gauge:	SN#	Cert Date:
13.96	PosiTector	232259	11/4/2026

HOLIDAY TEST

Conducted:	High Voltage	Low Voltage	Voltage
NA			

Gauge:		SN#			Cert Date:

Comments:

Apply 2-part consumers green to a final dft range of 10-16mils min.

Exp. Date:

Dura-Plate 235 A. 11/12/2028 B. 11/26/2027

Hi-Solids Poly: S. 4/26/2029 T. 4/22/2028

QTR Inspector: J. Brayer Nace Cert# 72547 Date: 7-17-26
Customer Inspector: Raja John Nace Cert# 011552 Date: 7-20-26
ACI